

Spatio-Temporal Predictive Modeling Techniques for Different Domains: a Survey

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Spatio-temporal prediction tasks play a crucial role in facilitating informed decision-making through anticipatory insights. By accurately predicting future outcomes, the ability to strategize, preemptively address risks, and minimize their potential impact is enhanced. The precision in forecasting spatial and temporal patterns holds significant potential for optimizing resource allocation, land utilization, and infrastructure development. While existing review and survey papers predominantly focus on specific forecasting domains such as intelligent transportation, urban planning, pandemics, disease prediction, climate and weather forecasting, environmental data prediction, and agricultural yield projection, limited attention has been devoted to comprehensive surveys encompassing multiple objects concurrently. This article addresses this gap by comprehensively analyzing techniques employed in traffic, pandemics, disease forecasting, climate and weather prediction, agricultural yield estimation, and environmental data prediction. Furthermore, it elucidates challenges inherent in spatio-temporal forecasting and outlines potential avenues for future research exploration.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence; Machine learning**; • **Information systems** → *Spatial-temporal systems*; • **Computer systems organization** → Robotics; • **Networks** → Network reliability;

Additional Key Words and Phrases: Agricultural Datasets, Transportation, Weather/Climate Datasets, Pandemic Datasets, Forecasting, Graph Neural Networks, Spatio-Temporal Data, Urban Computing, Transformers

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1 Introduction

Spatial-temporal or spatio-temporal data have several unique characteristics that distinguish them from other data types. One of the major characteristics of these data is their interdependence across both space and time simultaneously (Figure 1(a)) shows generalized spatial-temporal correlation/dependence of all regions across time steps). For example, traffic volume across different pairs of locations in a city (as shown in Figure 1(b)) shows different correlation patterns across different times. Spatio-temporal data are observed in several real-world applications that include the study of climate patterns, the spread of diseases, agricultural productivity, and urban development, to name a few. Thus, analyzing these data can help make informed decisions in various fields. However, data in several of these domains are often characterized by heterogeneity, sparsity, noise, high dimensionality, and large volume [61], all of which introduce additional challenges to the analysis task.

For example, spatio-temporal traffic data that may be used to analyze forecasting tasks such as traffic volume, traffic speed, traffic flow, and vehicular demand are collected using sensors and hence are prone to errors and missing values [203].

Similarly, climate data that may be used for modeling climatic conditions and weather forecasting poses unique challenges due to the complex and dynamic nature of the Earth's climate. Factors such as non-linear and chaotic behavior of the weather conditions, limited observational data, multiple interacting features, and extreme events pose further difficulty in weather predictions. Similarly, agricultural yield prediction involving the forecast of crop yields over a geographical area at a given period is difficult due to the complexity and variability of agricultural systems. Some other specific challenges in this domain include crop-specific factors, limited data availability, heterogeneity of farming practices, limited understanding of biophysical and socio-economic factors, multiple crops and crop rotations, and external factors such as weather conditions or other soil attributes. Pandemic and disease data for forecasting the impact and spread of diseases over a geographical area at a given time pose unique challenges due to the complex and dynamic nature of infectious diseases. Some specific challenges related to a pandemic and disease prediction include rapidly changing dynamics, limited data availability, multiple interacting factors, and uncertainty as well as a limited understanding of disease dynamics.

As a result, while the primary goal of spatio-temporal data analysis is to capture the long-term temporal patterns of the spatial data [48], these issues that are prevalent across various domains must be effectively addressed. To analyze and model spatio-temporal data, an interdisciplinary approach is, therefore, necessary, and to address pertinent issues, it is important to correctly comprehend the similarities and differences in the data features and the modeling techniques applied across many domains. For example, studying the data characteristics and methods used in traffic-related problems may provide vital cues to handle prediction problems in meteorological and climate domains. Furthermore, climate is a significant factor that affects crop production, and climate predictions can help farmers make informed decisions about planting and harvesting crops. Thus, predictions of temperature, rainfall, and other weather patterns can help farmers optimize their crop management strategies, including irrigation, fertilization, and pest control.

Thus, understanding these interdependencies can help researchers develop more comprehensive and integrated strategies for addressing the complex challenges in diverse domains. Existing surveys on spatio-temporal prediction often focus on specific domains, i.e., spatio-temporal traffic prediction and urban planning, agricultural yield prediction, pandemic or diseases prediction, and climate or weather prediction [80]. However, the literature lacks holistic surveys of the techniques across different domains for spatio-temporal data, which is essential to developing more effective techniques. The goal of this survey is to fill this gap. The major objective of this survey is to provide a comprehensive and interdisciplinary overview of state-of-the-art spatio-temporal

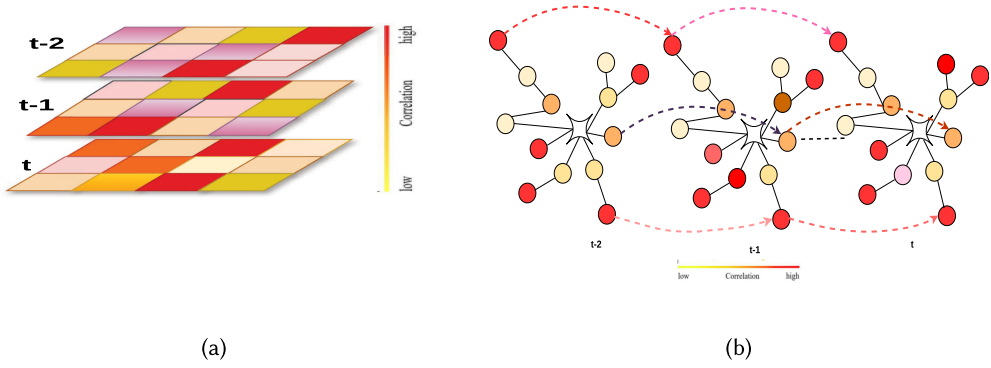


Fig. 1. (a): Spatial correlated regions/sensors changing with time $t-2$ to t (these characteristics are common in all spatial-temporal datasets), (b): Interdependent spatial correlation (represented by node colors) with target node (middle) and temporal correlation between each timestep $t-2$ to t (horizontal, shown for some selected nodes) in traffic data.

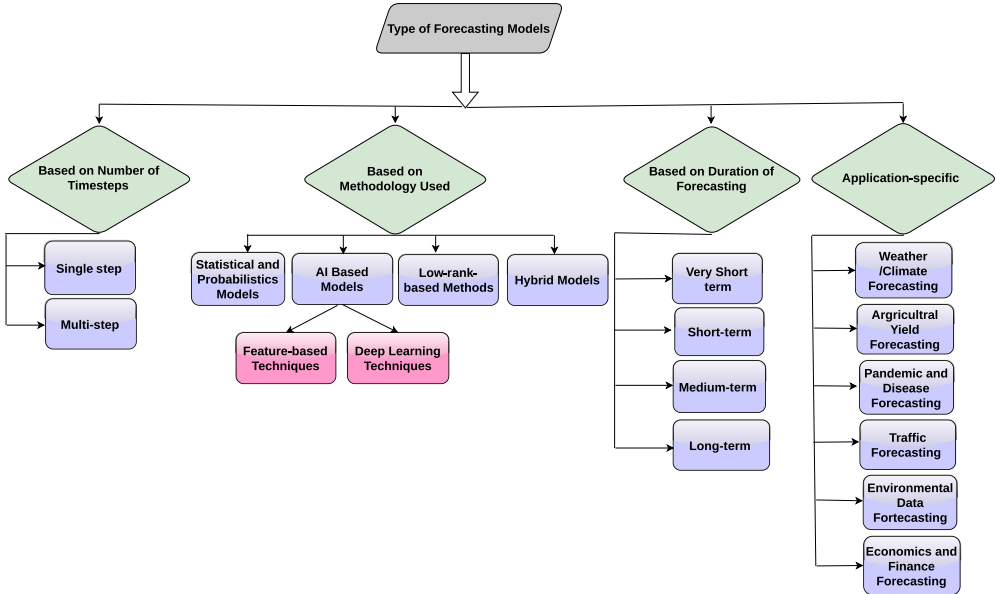


Fig. 2. All forecasting models. We include two categorizations in this survey article. Based on methodology 2. Application-specific.

prediction. The survey covers various domains and application areas, including climate modeling and prediction, transportation planning and traffic management, epidemiology and public health, agriculture, and food security. Furthermore, the survey aims to bridge these different fields and provide a comprehensive overview of the latest techniques and methods for spatio-temporal prediction (Figure 2).

The organization of this article is as follows: In Section 2, we provide a brief review of the existing relevant surveys across different domains and discuss their issues. In Section 3, we discuss our survey methodology and literature selection strategy. In Section 4, we discuss the importance of spatio-temporal data in various prediction tasks, and in Section 5, we subsequently explain

the challenges in dealing with spatio-temporal data jointly. Section 6 outlines the different public datasets used in the research across different domains. Section 7.1 provides an overview of the early time series-based techniques used in spatio-temporal modeling across different domains, whereas, in Section 7.2, we outline the feature engineering-based approaches used. In Section 7.3, we discuss the generative point process-based techniques, followed by the low-rank-based techniques that were proposed in Section 7.4. Section 7.5 outlines the major deep learning-based advances in spatio-temporal modeling. In Section 8, we discuss the practical application of spatial-temporal data modeling tasks and research results of various state-of-the-art models in each domain, followed by the conclusion in Section 9.

2 Related Works

In this section, we discuss the existing recent surveys in all domains and study the gaps that motivate the current article. A few appropriate surveys provide insight into the forecasting and modeling of spatial and temporal data. Each of these surveys takes a different approach to the literature, as will be explained in more detail below.

Hamdi, Atluri [10, 61] highlight the difficulties that researchers face when attempting to extract information from spatio-temporal data. They group these problems into distinct categories based on the data attributes, the tasks they have to do, and the application domain. Although these surveys extensively cover the goals and issues in spatio-temporal data mining, the models for spatio-temporal data forecasting are not well highlighted. Wang et al. examine the proposed works (up to 2019) in handling some spatiotemporal data modeling problems in several domains, providing more model-related information [182]. They defined the use of these models and divided the available literature into categories depending on the spatio-temporal data types, mining tasks, and DL models used. However, Shi et al. [157] focused on a systematic assessment of ML applications for spatio-temporal sequence forecasting. Both surveys identified the shortcomings of the examined works in their conclusions and offered ideas for new research directions. These surveys place a strong emphasis on predicting, but none of them address the three data-driven categories: time series-based, machine learning, and deep learning-based spatio-temporal forecasting models. In fact, the former, along with References [196] and [7], only featured the deep learning-based models, whereas the latter concentrated on models based on traditional machine learning. Certain reviews are focused on specific application domains. For example, Ermagun and Levinson [41] and Boukerche and Wang [20] reviewed the models used in traffic forecasting, while Tascikaraoglu [168] and Jin [80] reviewed the models used in smart city and urban development applications.

Existing surveys on spatial and temporal predictions above do not consider the differences and similarities across the domains. Here, we outline some of the limitations of the existing surveys.

Limited Scope: Many existing surveys focus on specific subfields or application areas of spatial and temporal prediction, such as climate modeling, agriculture yield modeling, or transportation planning [124, 162, 203]. While these surveys provide valuable insights into these areas, they may not capture the full range of challenges and opportunities involved in spatial and temporal prediction [10].

Limited Focus on Real-world Applications: Many surveys primarily focus on theoretical aspects and synthetic datasets. More emphasis could be placed on the challenges and successes of applying spatial-temporal prediction in real-world settings, where data quality, complexity, and external factors can significantly impact results [22, 162].

Lack of Interdisciplinary Perspective: Spatio-temporal prediction often involves knowledge from multiple disciplines, such as statistics, computer science, geography, and domain-specific expertise.

While some surveys may not fully integrate insights from these diverse fields, integrating such comprehensive reviews can encourage more robust cross-disciplinary approaches [80].

Lack of Updated Information: Many existing surveys were published several years ago and may not reflect recent advances in spatial and temporal prediction [51, 169, 182]. As the spatial and temporal prediction field rapidly evolves, it is essential to have up-to-date information on the latest techniques and methods.

All spatio-temporal models, including statistical, machine learning-based, and deep learning models, share similarities in modeling and forecasting temporal trends, patterns, and seasonality across domains such as traffic conditions, climate patterns, crop yields, and disease spread. Hence, through this survey, we attempt to synergize the spatio-temporal data forecasting models and techniques that are used across different domains, thus providing a more comprehensive study. Our survey has a broader scope, including discussions of various application areas and the latest techniques and methods. The survey can help identify research gaps, highlight emerging trends and challenges, and provide guidance on best practices for spatial and temporal prediction. This survey also mentions state-of-the-art datasets available in each above-mentioned domain, as well as all the recent research categorized as taxonomy.

3 Survey Methodology

In this section, we briefly outline our survey methodology. We start by exploring the current journals and conference articles dedicated to spatial-temporal prediction across diverse fields. These works help to delve into the latest advancements in spatial-temporal prediction models across various domains. (Figure 5(a) in Appendix A displays a word cloud visualization highlighting the most frequently encountered keywords in spatio-temporal research studies.) Our survey includes over 200 works (journals and conferences) related to spatial-temporal prediction and more than 15 survey papers (Figure 5(b), Appendix A). These references span a variety of publication types, such as journal articles, conference proceedings, books, and book chapters.

The majority of citations, accounting for 65 percent, come from journals published in well-regarded venues, followed by top-tier conferences, which constitute 35 percent, with the remainder being survey papers. Our research focused on high-impact journals including, *IEEE Transactions on Intelligent Transportation Systems*, *ACM Transactions on Intelligent Systems and Technology*, *IEEE Transactions on Knowledge and Data Engineering*, *Neurocomputing*, *IEEE Journal of Biomedical and Health Informatics*, *Precision Agriculture*, *Healthcare and Biological Sciences*, *Applied Energy*, *IEEE Transactions on Big Data*, *ACM Transactions on Database Systems*, *Cartography and Geographic Information Science*, and *Transactions in GIS*. We also reviewed contributions from prominent conferences such as the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), the IEEE International Conference on Data Engineering (ICDE), the IEEE Visualization Conference, the AAAI Conference on Artificial Intelligence, the International Joint Conference on Artificial Intelligence (IJCAI), International Conference on Machine Learning, Advances in Neural Information Processing Systems, and the International Conference on Ubiquitous Computing. Notably, 95 percent of the cited papers were published within the past five years.

4 Spatio-temporal Data and Their Importance in Different Applications

In this section, we outline how spatio-temporal properties are reflected in different applications and also study the importance of capturing these properties in different prediction-related tasks. We consider applications such as transport, urban planning, climate and meteorology, ecology, precision agriculture, and health and pandemics, to name a few.

Intelligent Transportation Systems: An **Intelligent Transport System (ITS)** is a collection of advanced technologies, communication systems, and data analysis techniques used to improve the safety, efficiency, and sustainability of transportation networks. ITS integrates various components and applications to enhance transportation management, optimize traffic flow, and provide travelers with real-time information [203]. ITS utilizes advanced sensors, including loop detectors, radar, and cameras, to collect real-time traffic data. This data includes traffic volume, speed, and occupancy information, which is used for traffic management, planning, and analysis purposes. Spatio-temporal data plays a vital role in an ITS by providing valuable insights into the dynamic nature of traffic and enabling effective decision-making and optimization of transportation systems. Traffic monitoring and analysis, traffic signal optimization, route guidance and navigation, transport planning and infrastructure design, and traffic prediction and forecasting are major tasks of intelligent transportation systems. Spatio-temporal properties play a crucial role in traffic prediction and forecasting, as they capture the dynamics of traffic flow over time and space. Spatio-temporal analysis helps identify recurring traffic flow patterns, such as rush-hour congestion or daily traffic fluctuations. Models can identify trends and forecast future traffic conditions by analyzing historical data. Congested locations can be identified at specific times using spatio-temporal data. By considering the temporal dimension, traffic prediction models can anticipate congestion-prone periods and provide insights for congestion mitigation strategies. Models can calculate travel times based on specific routes and the time of day by examining previous traffic patterns. Real-time navigation systems and route planning both benefit from this information. Additionally, spatio-temporal analysis aids in comprehending how traffic conditions alter in response to various occurrences or disturbances, such as collisions, construction, or special occasions.

Climate/Weather Prediction: Spatio-temporal climate or weather forecasting is a process that involves predicting future atmospheric conditions and climate patterns across both space and time. It combines the analysis of spatial data, which focuses on understanding weather patterns across different geographical areas, with temporal data, which examines changes over time. By integrating these two aspects, meteorologists and climatologists employ tools such as numerical weather prediction models, satellite imagery, Doppler radar, and climate models to make accurate predictions about weather phenomena and long-term climate changes. This forecasting helps in various sectors, such as agriculture, disaster management, transportation, energy production, and urban planning by providing valuable insights into the spatial and temporal distribution of weather events and climate trends. Spatio-temporal data for climate or weather forecasting is captured by deploying weather monitoring instruments, such as weather stations, satellites, radars, and buoys, strategically placed across different locations. These instruments measure various meteorological parameters such as temperature, humidity, wind speed and direction, precipitation, atmospheric pressure, and solar radiation [45]. The collected data is then integrated and analyzed to create comprehensive spatio-temporal datasets that capture the dynamics of weather and climate phenomena. This data is essential for understanding weather patterns, climate variability, and long-term climate trends, enabling the development of accurate weather forecasts, climate models, and informed decision-making in various sectors impacted by weather and climate conditions. The spatial variability of climatic and meteorological variables across various regions is better captured by spatio-temporal analysis. Understanding climate zones, microclimates, and regional weather patterns requires deep analysis of this data. The identification of temporal trends and climate and meteorological variable changes is made possible by spatial and temporal analysis. Scientists can identify patterns of climate change, such as rising temperatures, shifting precipitation patterns, or shifting weather seasons, by analyzing historical data across extensive periods. These tendencies aid in predicting future climatic scenarios and offer insights into long-term climate variations [43]. These circulation patterns impact many regional and temporal weather and climate regimes. For

describing weather systems such as low-pressure systems, high-pressure systems, cyclones, anticyclones, hurricanes, tornadoes, heatwaves, heavy rainfall events, and fronts, spatio-temporal analysis is crucial. These systems change throughout time and have diverse spatial features. Meteorologists can anticipate short-term weather, including these systems' movement, intensity, and associated meteorological conditions, by studying their spatio-temporal behavior.

Overall, spatio-temporal data is a critical tool for climate and weather prediction and is essential for developing effective strategies for mitigating the impacts of climate change and adapting to changing environmental conditions.

Precision Agricultural: Precision agriculture, also known as precision farming, is an innovative approach to agriculture that utilizes advanced technologies and data-driven techniques to optimize farming practices at a highly precise and localized level. By integrating tools such as GPS, remote sensing, and sensor technologies, precision agriculture enables farmers to collect and analyze data on soil conditions, weather patterns, and crop health. This information allows them to make informed decisions regarding planting, fertilization, irrigation, and pest management, tailored to the specific needs of different areas within a field. The goal is to maximize efficiency, minimize resource waste, reduce environmental impact, and improve overall productivity and profitability in agriculture. Agricultural yield prediction is a very essential task to achieve all of that. Spatio-temporal properties play a significant role in agricultural yield prediction, capturing the dynamic interactions between crops, climate, soil, and management practices across different spatial and temporal scales. The geographical variability of agricultural elements such as soil characteristics, topography, and nutrient availability is better captured by spatio-temporal analysis. Understanding the spatial variability in yield across various fields or locations depends heavily on this information. Scientists can identify climate patterns and their association with crop performance by analyzing historical climate data, including temperature, precipitation, solar radiation, and evapotranspiration. This knowledge aids in forecasting production variations under various climate scenarios and understanding how sensitive crops are to climatic influences. Models can forecast yield outcomes at various times during the growing season by considering the dynamic nature of crop growth and its response to environmental conditions. Scientists can assess the effects of various management practices on agricultural output by combining data on management inputs, time, and spatial distribution. Under particular management scenarios, this knowledge aids in predicting yield results and optimizing agricultural practices. The interactions between nearby fields or areas are also considered in spatio-temporal analysis. By combining spatial statistical approaches, yield prediction models can consider spatial interdependence. By analyzing the influence of neighboring fields on yield performance, models can improve the accuracy of predictions and account for spatially varying factors that impact yield. Historical yield data subjected to spatio-temporal analysis offers insights into long-term yield patterns and trends. Scientists can discover spatial and temporal trends, such as yield gaps, crop rotations, or the impact of particular climatic variables, by analyzing yield data across numerous years and locations. This information helps understand the drivers of yield variability and can contribute to more accurate yield predictions [124].

Pandemics and Diseases: Spatio-temporal pandemic and disease prediction use spatial and temporal data analysis to forecast and anticipate the spread of pandemics and diseases across different locations and time frames. This approach integrates various data sources, including epidemiological information, population density, mobility patterns, healthcare infrastructure, and environmental factors. By analyzing cases' spatial distribution and temporal evolution, researchers and public health authorities can identify patterns, trends, and hotspots, which can inform proactive measures and interventions. Spatio-temporal modeling techniques, such as epidemic models, geo-spatial mapping, and data-driven algorithms, help predict disease transmission dynamics, estimate future outbreak scenarios, and optimize resource allocation for prevention, surveillance, and control

strategies [66]. By analyzing spatio-temporal data, researchers can gain insights into the underlying patterns of disease transmission, distribution, and movement of infected individuals and develop more effective strategies for preventing and controlling the spread of infectious diseases [166]. By analyzing historical data on disease incidence, transmission rates, and other epidemiological factors, researchers can identify relationships and correlations that can be used to develop predictive models for future disease outbreaks.

In brief, spatio-temporal data is important for prediction tasks in various domains, as it provides context, captures dynamic behavior, identifies trends and patterns, models interactions and dependencies, enables data-driven modeling, supports decision-making, and facilitates monitoring and early warning systems. It plays a fundamental role in understanding and predicting complex spatio-temporal phenomena. There are challenges while modeling spatio-temporal data for various tasks. In the following section, we outline several challenges related to modeling spatio-temporal tasks.

5 Challenges in Dealing with Spatial and Temporal Task Jointly

Dealing with spatial and temporal prediction tasks jointly, which involves predicting patterns over space and time, poses several challenges due to these tasks' complex and interdependent nature (traffic prediction, climate/weather prediction, agricultural yield prediction, pandemic, and disease prediction). Here, we outline all the challenges of dealing with spatial and temporal tasks.

Data Availability and Quality: Spatio-temporal prediction tasks require many spatio-temporal data from multiple sources. Several variables, including measurement mistakes, sensor limits, collecting information processes, data processing methods, missing values, or any noise related to random variations might impact the data quality. The availability of spatio-temporal data can vary, depending on data ownership, licensing, privacy concerns, and data-sharing policies. Some data sources may be freely accessible, while others require specific permissions or subscriptions. So, these certain drawbacks lead to poor quality and affect the predictions [61].

Heterogeneity and Coupled Correlation in Data: Spatio-temporal data, with its inherent coupled correlation and heterogeneity, poses significant challenges to predictive modeling [59]. The complex interplay between spatial and temporal dependencies demands models that can capture intricate relationships across neighboring locations and past timesteps, while also accounting for the non-stationary nature of these correlations [25]. The heterogeneity of spatio-temporal data poses unique challenges stemming from diverse data sources, including sensors, satellite imagery, social media, and mobile devices, each with distinct formats, resolutions, and accuracies. This variety complicates integration efforts. Additionally, missing data, noise, and uncertainty inherent in measurements, further challenge reliable modeling, underscoring the importance of robust uncertainty quantification and management strategies.

Computational Complexity and Scalability: Combining spatial and temporal dimensions often results in very high-dimensional data, which can be computationally intensive and require specialized hardware or software [71]. Developing scalable algorithms and leveraging distributed computing are essential for handling large-scale spatio-temporal data.

Overfitting and Generalization: Complex models may risk overfitting to the training data, capturing noise rather than true underlying patterns [4]. Ensuring robust generalization to unseen data while avoiding overfitting is a key challenge in spatio-temporal prediction tasks.

Non-stationary Patterns in Data: Analysis and prediction have been severely restricted by non-stationary patterns in spatio-temporal data. Unlike stationary time series, which exhibit stable mean, variance, and autocorrelation structures, non-stationary time series display evolving patterns and trends [151]. These patterns can manifest as increasing or decreasing trends, seasonal

fluctuations, or other temporal dependencies. Environmental variations, human activity, or other dynamic elements can affect these patterns. Accurately capturing and modeling the shifting dynamics is difficult, since they need constant monitoring, discovering underlying causes, and a knowledge of complex relationships. Specialized methods that can adapt to changing trends and patterns while considering spatial and temporal dependencies are needed to deal with non-stationarity.

Incorporating External Factors: Spatio-temporal prediction tasks often require incorporating external factors such as weather data, demographic data, and socio-economic data, which can be difficult to integrate into existing models [71].

Interpretability and Transparency: Many factors make interpretability and transparency in spatio-temporal prediction difficult. Statistical methods, high-dimensional data, and clear mathematical algorithms can all contribute to the intrinsic complexity of spatio-temporal prediction models. It is challenging to comprehend and evaluate the underlying processes and correlations between variables in these models because of their complexity. Black-box techniques, such as deep learning or ensemble methods that put accuracy above interpretability, may be used in spatio-temporal prediction models. These models can make precise predictions but are ambiguous in how they do it. Without interpretability, it becomes difficult to comprehend the reasoning behind the predictions, identify the crucial variables, and evaluate the model's biases or limitations [169].

Uncertainty: The choice of modeling methods and parameters may lead to uncertainty. Predictions made by various models may be subject to varied degrees of uncertainty. When dealing with complex spatio-temporal data, choosing the appropriate model best for certain tasks and figuring out the best parameter choices can be difficult. This can make it difficult to make precise predictions, leading to prediction differences between models and scenarios [169].

Overall, it takes multiple types of cutting-edge methods from data preparation, machine learning, deep learning, spatial analysis, and domain-specific proficiency to tackle these problems. Providing efficient solutions for combined spatio-temporal prediction challenges also demands multi-disciplinary collaboration among researchers in spatial analysis, temporal modeling, and domain-specific expertise.

6 Datasets

Several spatio-temporal datasets may be used to represent how spatial and temporal dimensions interact in various domains. Researchers and professionals investigate, analyze, and create solutions for spatio-temporal challenges using these datasets as useful resources. Here, we outline and explain some important spatio-temporal datasets used in various spatio-temporal tasks and their respective features. Every dataset has different opportunities and problems for analysis and modeling. Using these statistics, researchers may progress in various areas, including environmental studies, public health, agricultural productivity for food supply, and urban planning and development. In Figure 3, we outlined all the categories of spatio-temporal data in different domains, such as intelligent transportation, urban planning and development, climate and weather, precision agriculture, and pandemics and diseases. In Tables 1, 2, 3, 4, we provide datasets for all domains with their links.

6.1 Intelligent Transportation Systems Datasets

GPS (Global Positioning System datasets): GPS trajectory datasets comprise sequences of geographical locations recorded over time by GPS devices, widely utilized in transportation research, urban planning, environmental monitoring, and more. Each entry typically includes latitude and longitude for geographic coordinates, a timestamp marking the recording time, and

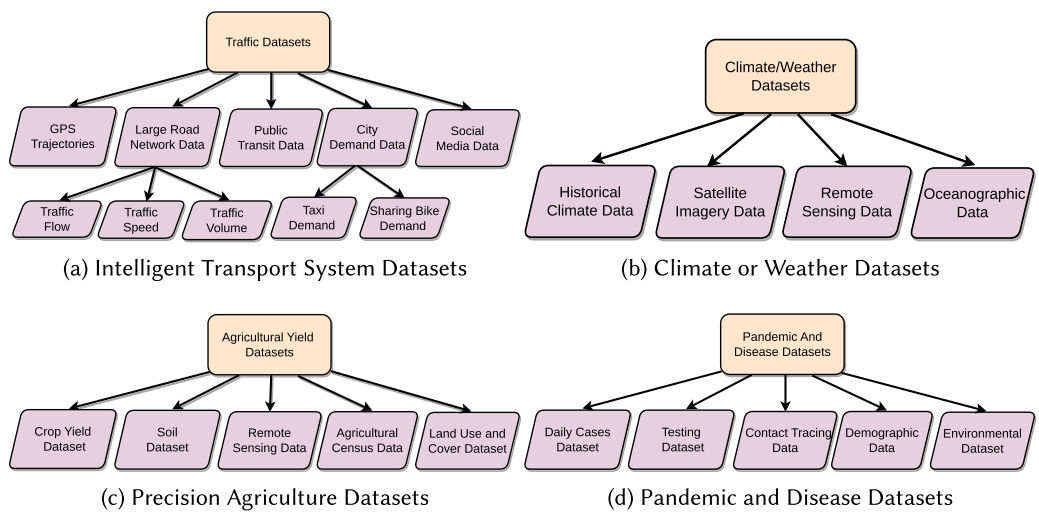


Fig. 3. All spatio-temporal datasets in different domains.

Table 1. Intelligent Transportation and Traffic Datasets

Dataset	Application	Location	Link
PeMSD4	Traffic Flow	California, USA	http://pems.dot.ca.gov/
PeMS-BAY	Traffic Speed	California, USA	http://pems.dot.ca.gov/
METR-LA	Traffic Speed	Los Angeles, USA	https://www.metro.net/
DRIVE Net	Traffic Data Platform	Seattle, USA	http://uwdrive.net/STARLab
Madrid city	Traffic Volume	Madrid, Spain	https://datos.madrid.es/portal/site/egob/
GAIA	Traffic Data Platform	Chengdu, Xian China	https://outreach.didichuxing.com/research/opendata/en/
GCM	Traffic Congestion, Speed	Gary, Chicago, Milwaukee, USA	http://www.travelmidwest.com/
Q-Traffic	Traffic Speed	Beijing, China	https://github.com/JingqingZ/BaiduTraffic-traffic-dataset
PortoTaxi	Taxi Demand	Porto, Portugal	https://www.kaggle.com/c/pkdd-15-predict-taxi-service-trajectory-i
Seattle Loop	Traffic Speed	Seattle, USA	https://github.com/zhilyong/Seattle-Loop-Data
NYC	Traffic Speed	New York, USA	https://www.kaggle.com/craaiatp/nyc-real-time-traffic-speed-data-feed
UDDS	Vehicle Speed	USA	https://www.epa.gov/emission-standards-reference-guide
OCC	Vehicle Volume, Speed	Los Angeles, USA	https://www.arb.ca.gov/regact/bus02/apcc.doc
HWFET	Vehicle Speed	USA	https://www.epa.gov/sites/default/files/2015-10/hwy10hrttable.txt
JN1015	Vehicle Multi Characteristics	Japan	https://www.epa.gov/sites/default/files/2015-10/jn1015col.txt
NYCC	Vehicle Speed	New York, USA	https://www.epa.gov/sites/default/files/2015-10/nycccol.txt
SC03	Vehicle Speed	USA	https://www.epa.gov/sites/default/files/2015-10/sc03col.txt
NEDC	Vehicle Multi Characteristics	Europe	https://github.com/dabo248/nedc
NGSIM	Traffic Data Platform	USA	https://data.transportation.gov
Minnesota Department of Transportation Traffic	Traffic flow, Traffic Speed	Twin Cities, MN, USA	http://data.dot.state.mn.us/datatools/
INRIX	Traffic Data Platform	USA	https://inrix.com/
Chicago Traffic Tracker	Traffic Speed	Chicago, USA	https://data.cityofchicago.org/Transportation/Chicago-Traffic-Tracker-Historical-Congestion-Esti/77hq-huss/data
US-Accidents	Traffic Accident	USA	https://smoosavi.org/datasets/us_accidents
Seattle-Loop-Data	Traffic Speed	Seattle, USA	https://github.com/zhilyong/Seattle-Loop-Data
Traffic Speed Guangzhou	Traffic Speed	Guangzhou, China	https://zenodo.org/record/120522
Traffic Flow Madrid	Traffic Flow	Madrid, Spain	http://datos.madrid.es
Traffic Flow Whitemud Drive	Traffic Flow	Whitemud Drive, Canada	http://www.openits.cn/openData1/700.html

may also feature altitude data, speed, bearing or heading, and accuracy levels that denote the confidence in the position measurements. These datasets can also contain a unique device ID for datasets involving multiple sources. GAIA, **OpenStreetMap (OSM)**, JN1015, NGSIM, and INRIX are a few examples of these types of datasets (Table 1).

Table 2. Climate Datasets

Dataset	Application	Temporal coverage Evaluation	Link
DHM	Precipitation	2001–2013, Daily	https://data.europa.eu/data/datasets/1cd18884-8677-4321-9daf-55b37845c9f0?locale=en
ERA-Int	Precipitation	2001–2013, 6-Hourly	https://apps.ecmwf.int/datasets/data/interim-full-mnth/levtype=sfc/
ERA5	precipitation	2001–2013, 1-Hourly	https://www.ecmwf.int/en/forecasts/datasets
ERA5-L	Precipitation	2001–2013, 1-Hourly	https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-land?tab=overview
HAR	Temperature, Precipitation, and Wind	2001–2013, Daily	https://www.klima.tu-berlin.de/index.php?show=daten_har
SST data	Sea_Surface Temperature	1985–2015	https://coastwatch.pfeg.noaa.gov/erddap
GSMaP	Rainfall	Latest	https://ghrc.nsstc.nasa.gov/home/content/gsmap-ifloods-dataset-now-available-ghrc
MODIS	Temperature	2002 to 2015	http://www.eorc.jaxa.jp/JASMES/index.html
Temperature and Humidity	Temperature and Humidity at Shenyang, Beijing, and Guangzhou	1 January 2015 to 31 December 2015	https://github.com/btblintellSense/Temperature-and-humidity-dataset

Table 3. Agricultural Yield Datasets

Dataset	Application	Temporal coverage Evaluation	Link
Crop Yield Prediction Dataset	Crop Yield, Temperature, Rainfall, Pesticides	1990–2016	https://www.kaggle.com/datasets/patelris/crop-yield-prediction-dataset
World Yield	Wheat-Yield	1961 to 2020	https://ourworldindata.org/crop-yields
GDHY	Yield Statistics and Satellite Data	1981–2016	http://www.earthstat.org/
Crop Yield World Wide	Crop Yield	2015	https://data.world/agriculture/crop-production
United States Department of Agriculture	Agricultural Yield	1992–2017	https://www.nass.usda.gov/Data_and_Statistics/

Table 4. Pandemic and Disease Datasets

Dataset	Application	Temporal coverage Evaluation	Link
COVID-19 in Africa	COVID-19	14 February to 15 April 2020	https://covid19.who.int
Johns Hopkins University(JHU) Coronavirus Resource Center	Covid various parameters	March 22, 2020 to June 10, 2020	https://github.com/CSSEGISandData/COVID-19
IQVIA's claims data, precipitation	Patient claims data and Prescription data	March 22, 2020 to June 10, 2020	https://github.com/CSSEGISandData/COVID-19
The Soho Outbreak	Cholera cases and deaths	19th Century	https://geodacenter.github.io/data-and-lab/snow/
COVID-19	Global COVID-19 cases all parameters	May 3, 2020 to April 23, 2021	https://covid19.who.int/
Sierra Leone Ministry of Health and Sanitation	Ebola and cholera	May 2014 to September 2015	https://github.com/rek160/Sierra-Leone-Cholera-Ebola
COVID-19 cases from Mexico	COVID-19	2020–21	https://www.kaggle.com/marianarfranklin/mexico-covid19-clinical-data/metadeta

Large Road Network Datasets: These datasets are comprehensive collections of geospatial information used to represent road systems for various applications, including urban planning, traffic management, navigation, and environmental analysis. These datasets contain detailed information about nodes (intersections or endpoints or sensors) and edges (road segments), complete with attributes such as road type, directionality, speed limits, and connectivity. They also include details on infrastructure such as bike lanes and public transport routes. Essential for GPS-based navigation systems, these datasets support traffic flow optimization, urban development, and transportation simulations. METR-LA, PEMS-BAY, PEMS03, and PEMS07 are a few examples of these datasets (Table 1).

Public Transit Datasets: Public transit datasets encapsulate detailed aspects of public transportation systems, including bus, tram, subway, and rail networks. They typically feature extensive data on routes, schedules, real-time vehicle locations, ridership patterns, and specific attributes of stops and vehicles. These datasets are instrumental for transportation planners, developers, and researchers in optimizing transit routes, scheduling services, enhancing passenger information services, and conducting performance analysis. Adelaide Metropolitan transport data, NGSIM, and NEDC are a few examples of this type of dataset (Table 1).

City Demand Datasets: City demand datasets are critical for analyzing and predicting urban needs such as traffic, energy use, and public service consumption. These datasets encompass a range of

data from specific cities, covering various time intervals to reveal temporal patterns, and are often segmented by geographic regions. Common features include temporal markers (such as timestamps and seasonality), spatial data (including Geo-location and region-specific attributes), and demand-specific metrics (such as traffic counts and energy consumption). Porto Taxi, NYC, and Chicago traffic trackers are examples of this dataset type (Table 1).

6.2 Climate/Weather Datasets

Historical Climate Data: These datasets comprise detailed records of atmospheric conditions collected over time from weather stations worldwide. Key features include temperature, precipitation levels, atmospheric pressure, humidity, wind speed and direction, sunshine duration, solar radiation, cloud cover, and visibility. This data is essential for studying climate patterns, developing climate models, and understanding long-term climate changes. It supports research, agriculture, environmental planning, and policy-making by enabling accurate weather forecasting. ERA-Int, ERA5, and HAR are examples of this dataset type (Table 2).

Satellite Imagery Data: Satellite imagery data is crucial for spatial-temporal climate prediction, providing high-resolution, multi-spectral observations of Earth's surface and atmosphere. Features of this data include spatial resolution (detail level), temporal resolution (frequency of observations), spectral bands (variety of wavelengths), radiometric resolution, and extensive coverage, which is especially beneficial for remote areas. This data supports various applications such as weather forecasting, climate modeling, environmental management, and disaster response. GSMaP and MODIS are examples of this dataset type (Table 2).

Remote Sensing Data: Remote sensing data, derived from satellite and aerial sensors, plays a crucial role in spatial-temporal climate/weather prediction by providing comprehensive, multispectral observations of Earth's surface and atmosphere. This data features various resolutions—spectral, spatial, temporal, and radiometric—each critical for analyzing different environmental and atmospheric phenomena. High spatial resolution helps in detailed observation of specific events like deforestation, while high temporal resolution is essential for tracking dynamic climate processes. MODIS, NDVI, and **Enhanced Vegetation Index (EVI)** are a few examples of Remote sensing data (Table 2).

Oceanographic Datasets: Oceanographic datasets are vital for spatial-temporal climate/weather prediction, offering extensive insights into the oceans' physical, chemical, and biological characteristics. These datasets include measurements of ocean temperature, salinity, sea levels, wave and tide dynamics, chemical properties, and biological data such as marine biodiversity. **Sea surface temperature (SST)** data (Table 2) serves as an example of this type of information. Such detailed information is crucial for enhancing climate models, improving weather forecasts in coastal regions, and monitoring environmental changes such as ocean acidification and marine ecosystem health.

6.3 Precision Agricultural Datasets

Crop Yield Datasets: Crop yield datasets are crucial for predicting agricultural productivity and managing food resources effectively. These datasets provide detailed information on yield measurements, geographic and temporal data, types of crops, environmental conditions, and cultivation practices. Additional insights from remote sensing data enhance the precision of these datasets. These datasets are utilized in yield prediction, agricultural planning, and policy-making for food security (Table 3).

Soil Datasets, Remote Sensing Data, and Agricultural Census Data: Soil datasets, remote sensing data, and agricultural census data are integral to spatial-temporal agricultural produce prediction, each

offering unique insights that enhance farming efficiency and crop yield forecasting. Soil datasets detail critical properties such as type, nutrient levels, pH, moisture, and organic content, which are essential for tailored agricultural practices. Remote sensing data provides high-resolution, multi-spectral imagery of agricultural lands, enabling continuous monitoring of crop health and environmental conditions. Agricultural census data includes comprehensive statistics on farm demographics, cropping patterns, inputs, outputs, and economic factors, which are crucial for understanding broader agricultural trends and making informed policy decisions. Together, these datasets empower precision agriculture by providing the necessary tools for optimal resource management, timely interventions, and strategic agricultural planning.

6.4 Pandemic and Disease Datasets

Contact Tracing Data, Demographic Data, and Environmental Data: These datasets are fundamental components of spatial-temporal pandemic prediction, offering critical insights into disease transmission dynamics and informing effective mitigation strategies. Contact tracing data tracks interactions between infected individuals and their contacts, providing valuable information on transmission routes and patterns. Demographic data encompasses socio-economic and population characteristics influencing disease susceptibility and healthcare access. Environmental data includes factors such as climate conditions, air quality, and geographical features, which impact disease transmission and persistence (Table 4).

Global Health Data from WHO: Some important features of these datasets are geographic locations, timestamps (dates of reported cases, deaths, and recoveries), disease metrics (confirmed cases, deaths, recoveries), demographic information (age, gender, and sometimes race/ethnicity of affected individuals), and health infrastructure data.

7 Spatio-temporal Prediction Methods

Spatio-temporal prediction techniques are employed to forecast future values or states of variables that exhibit both spatial and temporal dependencies. These methods are commonly used in all the above-discussed spatio-temporal prediction tasks [33, 78, 152]. The selection of the prediction method depends on the specific characteristics of the problem, available data, computational resources, and desired level of accuracy. Different methods may be more suitable for different scenarios. Spatio-temporal prediction methods can be broadly divided into several categories based on underlying principles and techniques (Figure 4). Here, we will explain several popular spatio-temporal prediction methods. In Tables 5, 6, 7, 8, we outline all the state-of-the-art work done in all the domains, i.e., intelligent transportation, climate or weather, precision agriculture, and pandemic and health.

7.1 Time Series-based Methods

Time series models are specialized in capturing temporal dependencies within data, which is crucial for forecasting and understanding time-related patterns. First, **Autoregressive Integrated Moving Average (ARIMA)** [57] models autocorrelation, trend, and seasonality, while **Seasonal ARIMA (SARIMA)** captures seasonal components within data. ARIMAX expands ARIMA by incorporating exogenous factors. **Space-Time ARIMA (ST-ARIMA)** extends [217] SARIMA to include spatial and temporal dependencies, modeling complex interactions. **Vector Autoregression (VAR)** models multiple time series simultaneously, capturing dependencies between variables. VARIMA extends VAR by adding integration components, handling non-stationarity, and capturing short- and long-term dynamics. However, the time series must be stationary for the ARIMA, SARIMA, and variant models, so its statistical characteristics must remain constant across time.

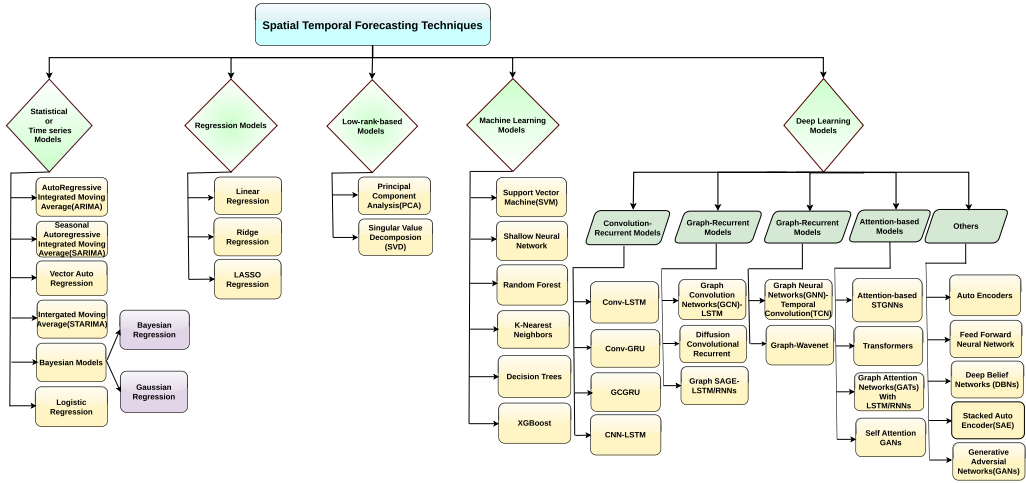


Fig. 4. Different techniques for predicting spatio-temporal data.

These models require stationary data and struggle with non-linearities, outliers, and complex interactions. Kalman filters excel in real-time state estimation from noisy data, used in forecasts such as traffic, weather, pandemic, and precision agriculture [3]. However, due to the failure to process various variables and fixed assumptions, statistical techniques are too straightforward to derive dynamic spatio-temporal patterns. Conventional machine learning methods are utilized in various spatio-temporal forecasts to handle high-dimensional data and derive intricate traffic patterns.

7.2 Traditional Machine Learning-based Methods

Machine Learning (ML)-based models for spatio-temporal data modeling provide the opportunity to extract meaningful and informative representations from the raw data. It enhances model performance, incorporates contextual information, reduces dimensionality, improves interpretability, handles missing data and outliers, and allows for the integration of external data sources. Together, these advantages support the creation of reliable and accurate spatio-temporal models. Machine learning enables the development of forecasting models considering vast quantities of diverse data from various sources. Numerous studies have been conducted on using ML algorithms for various spatio-temporal predictions. This section discusses the most common machine learning techniques for spatio-temporal data modeling.

Support Vector Machines (SVM): SVM utilizes a linear model in feature space, typically derived from transforming the input space via a basis function [159]. It optimizes a linear separation boundary concerning support vectors, solving a convex quadratic programming problem to ensure a global minimum. SVM's flexibility in generating non-linear boundaries is facilitated by kernel functions that transform inseparable input data into separable feature space data, although selecting an accurate kernel remains challenging. SVM finds frequent application in modeling spatio-temporal features across diverse networks, such as metropolitan and rural areas. Standard kernels such as Gaussian radial basis and polynomials are commonly used but may require customization for specific tasks, such as wavelet kernels for non-stationary traffic speed characteristics. In various spatio-temporal prediction tasks, SVM has been applied to weather forecasting, atmospheric temperature prediction, and COVID-19 case predictions worldwide [170]. However, SVMs may struggle with changing data distributions over time, limiting their effectiveness in capturing non-stationary patterns. They also face challenges in modeling complex spatial

Table 5. Traffic Prediction Papers

Model Type	Task	Spatial and Temporal Dependency Modeling	References
Probabilistic Graph Methods and Low-rank-based Models	Traffic Speed	Probabilistic Graph	Furtlehner et al. [49]
	Traffic Flow	Low Rank Dynamic Mode Decomposition (LRDMD)	Yu et al. [202]
	Traffic Speed	DMD based low rank tensor prediction model (DMD-LRT)	Zhang et al. [210]
Statistical or Time-series-based Models	Traffic Speed	- , Kalman Filters (KF)	Lippi et al. [108]
	Traffic Speed	-, SARIMA, ARIMA	Williams and Hoel [188]
Traditional Machine Learning-based Models	Traffic Congestion	Support Vector Machine (SVM), Support Vector Machine (SVM)	Ata et al. [9]
	Traffic Speed	-, Artificial Neural Network (ANN)	Tang et al.
Deep Learning-based Methods	Traffic Speed	-, Gated Recurrent Unit (GRU)	Zhao et al. [213]
	Traffic Speed	-, Gated CNN	Liu et al. [112]
	Traffic Speed	SGCN - CNN	Diao et al. [36]
	Traffic Speed	CapsNet , -	Kim et al. [93]
	Traffic Speed	-, GRU + attention	Wu et al. [189]
	Traffic Speed	-, Seq2Seq (RNN)-Attention	Zhang et al. [205]
	Travel Time	Graph Convolution (GCN) - Recurrent Neural Network (RNN)	Katayama et al. [86]
	Traffic Speed	Attention + GCN - Recurrent Neural Network (RNN)	Huang et al. [67]
	Traffic Speed	-, TCN	Zhao et al. [215]
	Traffic Speed	Attention - Attention	Park et al. [132]
	Traffic Speed	Graph Attention (GAT) + Temporal Convolution (TCN) - (GAT) + (TCN)	Zhang et al. [208]
	Traffic Flow	Attention + Convolution - Attention + convolution	Guo et al. [58]
	Metro Crowd Flow and Road network Traffic Flow	Multi-Head Attention - Multi-Head Attention	Zhu et al. [219]
	Metro Crowd Flow, Bus passenger flow, Traffic Speed and Occupancy	Modified GCN-	Zhang et al. [209]
	Traffic Speed	GCN - GRU	Zheng et al. [218]
	Traffic Volume	GCN - GAT	Shang et al. [150]
	Metro Crowd Flow and Traffic Speed	Attention - Attention	Wang et al. [176]
	Traffic Flow	Attention and Dilated GCN - Attention and TCN	Wang et al. [185]
	Traffic Congestion	CNN - Vision Transformer	Ramana et al. [138]
	Traffic Flow	GCN - Bi-directional GRU	Jiang et al. [77]
	Traffic Flow	GCN - TCN	Wu et al. [191]
	Bus Travel Demand	GCN - LSTM	Zhao et al. [214]
	Traffic Flow	GCN - GLUs	Jin et al. [79]
	Uber Order	CNN - Adaptive GCN	Jin et al. [81]
	Traffic Congestion	Transformer - GCN	Jin et al. [80]
	Traffic Flow	Hybrid Transformer	Liu et al. [110]
	Traffic Flow	Adaptive Transformer	Wang et al. [181]
	Traffic Flow	Transformer Model	Luo et al. [116]

dependencies and temporal dynamics due to their primary focus on class separation via hyperplanes. Moreover, SVMs may become computationally demanding for large-scale spatio-temporal problems as dataset sizes increase. Therefore, while SVMs offer robust classification capabilities, particularly in structured data, their limitations in handling dynamic spatio-temporal patterns and scalability issues should be considered.

Random Forest (RF): RF is an ensemble algorithm that aggregates predictions from multiple decision trees trained on different bootstrap samples of the dataset [135]. It ensures diversity among trees by randomly selecting subsets of attributes for each split. This method is effective for spatio-temporal predictions, leveraging feature engineering to capture spatial and temporal patterns. Variables such as geographic coordinates, climate data, and road conditions are crucial inputs for weather and traffic forecasting models. In precision agriculture, Random Forests are used to optimize water usage and improve harvest outcomes. However, interpreting Random Forests can be challenging due to their complex ensemble nature.

K-Nearest Neighbors (KNN): KNN is a versatile machine learning algorithm used for spatio-temporal forecasting tasks [63]. It leverages feature similarity to predict future values, performing well in various applications such as predicting rainfall and analyzing pandemic spread [19].

Table 6. Climate/Weather Prediction Papers

Model Type	Model	Parameters Used	References
Probabilistic Graph Models and Low-rank-based Models	Probabilistic Approach based on Fuzzy Bayesian network (FBN)	Air Temperature, Pressure, Humidity	Das et al. [31]
	Probabilistic Markov model (PMM)	Weather Elements	Kaneriya et al. [84]
Statistical or Time-series-based Models	ARIMA	Wind Speed	Tarade et al. [167]
	ARMA, VAR	Wind Speed	Erdem et al. [40]
Machine Learning-based Models	ANN	Wind speed, Rainfall	Tran et al. [171]
	Particle Swarm Optimization (PSO) and SVM	Temperature	Du et al. [39]
	Random Forest	Temperature	Singh et al. [158]
	Support Vector Machine	Rainfall	Rasel et al. [139]
Deep Learning-based Models	Recurrent Neural Network (RNN)	Temperature, Wind Speed, Air pressure, Humidity	Salman et al. [146]
	Deep Neural Network (DNN)	Temperature, Wind Speed, Air Pressure, Humidity, Wind Direction	Ghoneim and Manjunatha [55]
	SDAE	Air Pressure, Temperature, Humidity, Wind speed	Hossain et al. [64]
	DBNPF	Wind Speed, Humidity, Temperature, Air Pressure, Precipitation	Zhang et al. [207]
	Stacked-LSTM	Wind Speed	Liang et al. [107]
	Wavelet Packet-ANN	Wind Speed	Liu et al. [109]
	Radial Basis Function (RBF) and Multi-layer perceptron (MLP)	Wind Speed, Temperature	Saba et al. [145]
	SAE-LSTM	Temperature, Wind Speed	Wu et al. [190]
	Wavelet-SVR	Wind Speed	Dhiman et al. [35]
	3-D Neural Network	Weather forecasting	Bi et al. [18]
	Deep generative model	Precipitation	Zhang et al. [211]
	OSORELM and hybrid mode decomposition (HMD) method	Wind Speed	Zhang et al. [204]
	GRU based Deep Larning Model	Temperature, Relative Humidity, Wind, and Radar Echo	Yu et al. [201]
	Attention	Air Temperature	Zhao et al. [216]
	Space-time Transformer	Earth System	Gao et al. [52]
	Transformer with Adaptive Fourier Neural Operator	Surface wind speed, Precipitation, and Atmospheric water vapor	Pathak et al. [133]

Table 7. Agricultural Yield Prediction Papers

Model Type	Models	Parameters and Data Used	References
Probabilistic Graph Methods and Low-rank-based Models	PCA	Pigeon Pea Yield	Pandey et al. [131]
	Hidden Markov Chain	Crop Yield	De et al. [32]
Statistical or Time-series-based Models	ARIMA	Rice Yield	Reddy et al. [141]
	ARMA, SARIMA, and ARMAX models	Crop Yield	Bang et al. [14]
Machine Learning-based Models	Deep Gaussian Process Model	Soybean Yield	You et al. [197]
	Gradient Boosting, Support Vector Regression, K-Nearest Neighbors	Wheat, Spring barley, Sunflower, Sugar beet, Potatoes Yield	Paudel et al. [134]
	Random Forest	Apple Yield	Bai et al. [12]
	Logistic Regression	Apple Yield	Aggelopoulou et al. [2]
	XGBoost	Blueberry Yield	Obsie et al. [125]
	Bayesian Neural Networks (BNN)	Crop Yield Data	Johnson et al. [82]
	Machine Leaning Hybrid Models	Tea Crop Yield	Batool et al. [15]
	Machine Learning Ensemble Models	Wheat Yield	Fei et al. [46]
Deep Learning-based Models	LSTM	Wheat Yield Data, Surface reflectance, NDVI, Humidity	Wang et al. [184]
	3D-CNN	Surface Reflectance, Land Cover and Surface temperature	Gavahi et al. [53]
	CNN-LSTM	Land Surface Temperature and Reflectance, Soybean Crop Yield	Sun et al. [163]
	Deep Neural Network	Maize Yield	Khaki et al. [88]
	Deep Transfer Learning	Soybean Crop Yields	Wang et al. [175]
	GNN-RNN	Corn and Soybean Yield	Fan et al [44]
	CNN-DNN	Soybean Yield	Oikonomidis et al. [126]
	Stacked LSTM	Rice Yield	Geetha et al. [54]
	Bi-LSTM	Rice Yield	Saveetha et al. [148]
	Transformer Model	Wheat yield prediction	Junankar et al. [83]
	Transformer Model	Multiple crop yiled	Ibanez et al. [70]
	Transformer Model	Sugercane yield Prediction	Krishman et al.[94]

Table 8. Pandemics and Disease Prediction Papers

Model Type	Model	Parameters and Data Used	References
Probabilistic Graph Methods and Low-rank-based Models	Markov Model	Covid-19 in India	Arumugam et al. [6]
	Markov chain Monte Carlo techniques	Dengue Fever Prediction	Martheshwaran et al. [119]
Statistical or Time-series-based models	ARIMA and Prophet	Covid-19 in World	Naresh et al. [95]
	ARIMA and NAR	Covid-19 in India	Khan et al. [90]
	ARIMA, AR, MA	Covid-19 in India	Kumar et al. [96]
	Fuzzy clustering, Kalman Filter	Covid-19 Daily Deaths in Brazil	dos Santos et al. [38]
	ARIMA	Total Cases of Covid-19	Shahid et al. [149]
	Statistical Models	Dengue in Indonesia	Jaya et al. [73]
Machine Learning-based Models	ARIMA, Prophet	Covid-19 Cases Per Day From J.H.University, World Weather Page, Wikipedia	Devaraj et al. [34]
	RF, DT, ANN, CR	Clinical Data, Patients demographics	Assaf et al. [8]
	DT, LoR, RF, ExtraTrees, NB, KNN	Routine Blood Exams	Brinati et al. [21]
	LR, PR	Covid-19 Cases and Deaths Per Day	Chakraborty et al. [26]
	Machine Learning Models	Covid-19 in Many Countries	Malki et al. [117]
	Hawkes Point Process Model	Ebola in Congo	Kelly et al. [87]
Deep Learning-based Model	LASSO SVM, LR, ES	Covid-19 World Wide Patients	Rustam et al. [144]
	LSTM	The Number of Covid-19 Cases in Canada (Time Series Data)	Chimmula et al. [30]
	EEMD, ANN	Covid-19 Total Cases, Deaths and Recovery	Hasan et al. [62]
	GRU, SVR, Bi-LSTM, LSTM	Covid-19 Cases	Shahid et al. [149]
	Stacked LSTM, Conv LSTM, Bi-directional LSTM	Total Cases of Covid-19 in India and USA	Shastri et al. [155]
	Deep-LSTM Ensemble Model	Total Cases and Death of Covid-19 in India	Shastri et al. [156]
	Transfer Graph Neural Networks	Covid-19 in Spain, Italy, England and France	Panagopoulos et al. [130]
	Attention	Covid-19 in USA	Gao et al. [50]
	Attention	Covid-19	Li et al. [102]
	Deep Learning models	Dengue in Brazil	Li et al. [105]
	Transformer model	Seasonal Influenza	Li et al. [101]
	Transformer model	Covid-19	Banerjee et al. [13]
	Transformer Model	Infectious Disease	Chen et al. [27]

However, an important consideration in KNN is selecting an appropriate distance metric—common choices include Euclidean or Manhattan distance, tailored to capture both spatial and temporal aspects. Despite its effectiveness, KNN’s sensitivity to the parameter k can impact performance. Choosing an optimal k is crucial to balance between overfitting with a small k and introducing bias with a large k , ensuring accurate predictions without compromising on local patterns.

Gaussian Process Method (GP): GP is a powerful non-parametric model widely used for spatio-temporal data modeling and forecasting [220]. It excels in capturing complex relationships in data due to its flexible and probabilistic framework. The GP method is particularly suitable for spatio-temporal forecasting, as it effectively models spatial and temporal dependencies using covariance functions (kernels) such as squared exponential, Matérn, and periodic kernels. These functions specify relationships between data points based on observed data characteristics. In practice, GP involves fitting the model by estimating hyper-parameters of the covariance function through methods such as maximum likelihood estimation or Bayesian inference. This step determines the model’s smoothness, length scales, and periodicity. The predictions are then generated based on these parameters. Applications of GP in spatio-temporal tasks include predicting barley yields and wind power, demonstrating superior performance compared to other machine learning techniques [153]. Despite its effectiveness, implementing GP can be computationally intensive, especially with large datasets. Techniques like the Sparse Gaussian Process help mitigate scalability issues by approximating the covariance matrix. However, GP’s main drawbacks include its computational complexity, requiring quadratic storage and cubic learning calculations, which can limit its applicability in certain scenarios.

Shallow Neural Network: Due to its high robustness, the **Shallow Artificial Neural Network (Shallow ANN)**, modeled on the concept of the neurons in the biological brain and consists of an extensive network of neurons (computational cells), has become a common technique in the spatio-temporal forecast. A Shallow ANN was used in research to forecast the different spatio-temporal conditions of transportation, weather, agricultural yield, and pandemic predictions [42]. To determine the impact of the weather component, researchers Huang et al. [68] input the unfavorable weather information into Shallow ANN.

The lack of interpretability and explainability in machine learning models, particularly deep neural networks, is a significant drawback when modeling spatio-temporal data. These models are often considered “black-box” due to their complex decision-making processes, making it challenging to understand underlying relationships and interpret results. Additionally, these models require substantial labeled training data to generalize effectively, which can be costly and difficult to obtain for specialized or rare cases. They may also struggle to generalize to new spatio-temporal patterns that differ significantly from the training data distribution. Choosing and tuning models and hyperparameters can further add to the complexity and time investment. These challenges underscore the need to carefully consider model suitability and assessment in specific spatio-temporal modeling scenarios. So, in recent days, different deep learning-based techniques have drawn the attention of researchers who are modeling spatial and temporal data.

7.3 Probabilistic Graph Methods

Probabilistic Graph Methods offer a structured approach to modeling uncertainty in spatio-temporal data by representing and propagating probabilistic distributions. They handle uncertainties arising from measurement errors, missing data, and inherent process variability. Bayesian networks and Markov random fields graphically depict dependencies among variables, where nodes represent variables and edges signify probabilistic relationships. **Static Bayesian Networks (BN)** [174] use directed acyclic graphs, while Markov Fields employs undirected acyclic graphs. **Hidden Markov Models (HMM)** extend these concepts to dynamic Bayesian networks, effectively modeling spatio-temporal unpredictabilities [92]. BNs accommodate both discrete and continuous variables, offering flexibility in representation. HMMs have been applied to tasks such as traffic speed forecasting, demonstrating resilience to noisy data by characterizing dynamic speed changes. **Fuzzy Bayesian networks (FBN)**, introduced by Das et al. [31], predict weather by forecasting spatio-temporal correlations among climate variables. When choosing between probabilistic models and machine learning for spatio-temporal data, considerations include analysis goals, data availability, interpretability needs, and the balance between capturing uncertainty and achieving prediction accuracy.

7.4 Low-rank-based Methods

Low-rank-based methods model spatio-temporal data by reducing high-dimensional data into lower-dimensional structures, such as principal components or low-rank matrices. This approach captures dominant spatial and temporal patterns, mitigating the curse of dimensionality, enhancing computational efficiency, and aiding data analysis and interpretation. These methods are robust against missing data and noise, focusing on underlying structures rather than individual measurements. They are especially useful when data exhibit strong correlations and redundancies, providing a concise representation that retains essential information.

Next, we outline some of the low-rank-based techniques used for spatio-temporal prediction.

Singular Value Decomposition (SVD) is a matrix factorization technique used in spatio-temporal forecasting, decomposing a dataset into a left singular matrix, diagonal singular values matrix, and right singular matrix [76]. It identifies spatial and temporal patterns, which are

crucial for forecasting by projecting data onto these patterns. **Singular Spectrum Analysis (SSA)** extends SVD's utility by decomposing time series into spatio-temporal components for tasks such as weather forecasting, stock market prediction, crop yield prediction, and traffic prediction. Ji et al. [76] applied SVD to climate data for effective prediction, while Miao et al. [120] introduced ESARIMA integrating SVD with empirical mode decomposition to enhance traffic flow prediction. However, SVD's linear assumptions may limit accuracy in capturing complex, nonlinear spatio-temporal dependencies, potentially affecting performance, especially with large-scale datasets and memory constraints.

Principal Component Analysis (PCA) is a matrix factorization technique beneficial for spatio-temporal forecasting [122]. Unlike SVD, PCA determines components based on variance maximization rather than singular values. It identifies orthogonal components capturing data patterns, useful for forecasting. PCA reveals large-scale spatial trends in early components and finer local details in subsequent ones. It is utilized by Skittides et al. [160] for wind speed prediction. Guorong et al. [60] propose a hybrid PCA-Wavelet Neural Network for traffic forecasting. Yet, PCA may lose nuanced data complexity and assume linear relationships, limiting suitability for nonlinear spatio-temporal datasets.

One major drawback of low-rank-based methods is their limited ability to capture complex nonlinear relationships. Low-rank models assume linearity and represent the data as a low-rank matrix or a combination of low-rank matrices, which may not adequately capture the intricate spatio-temporal patterns present in the data. To mitigate these limitations, more advanced techniques that can handle nonlinearity, heterogeneity, and missing data, as well as provide interpretable representations, such as deep learning models or spatio-temporal graph-based methods, may be more suitable.

7.5 Deep Learning-based Methods

Deep learning has several advantages over traditional machine learning approaches for spatio-temporal data modeling. One key advantage is the ability of deep learning models to learn meaningful features from raw data automatically. This eliminates the need for manual feature engineering, which can be time-consuming and require domain expertise. Deep learning models excel at capturing complex spatio-temporal patterns by leveraging multiple layers of nonlinear transformations. They can handle high-dimensional data and capture intricate relationships, making them well-suited for modeling spatio-temporal dynamics. Additionally, deep learning models can effectively handle large-scale datasets and benefit from parallel processing and GPU acceleration, enabling efficient training and inference. Overall, deep learning-based models can derive the complicated pattern between numerous spatio-temporal variables and various spatio-temporal prediction conditions (traffic speed/volume/flow, climate/weather prediction, agricultural yield, pandemic prediction).

The holistic integration of spatial and temporal features is a pivotal pursuit in enhancing the predictive prowess of spatio-temporal models. A prevalent approach, frequently witnessed in the literature, involves the separate extraction of spatial and temporal features followed by their aggregation, often through techniques such as concatenation, linear transformation, and attention mechanisms [193]. While convolutional methodologies have proven their mettle in spatial dependency modeling, sequential models have gained prominence in capturing temporal dependencies. Notably, the most widely employed technique for capturing spatio-temporal interplay revolves around amalgamating distinct feature extraction techniques. In particular, the fusion of **Recurrent Neural Networks (RNNs)** and **Graph Convolutional Networks (GCNs)** has garnered considerable attention for its efficacy in seamlessly unifying spatial and temporal dimensions [183]. Yet, the landscape of spatio-temporal modeling continues to evolve,

witnessing the emergence of innovative methodologies capable of jointly encapsulating spatial and temporal dependencies. These approaches show potential for modeling complicated data interdependencies, where changes in one dimension impact other attributes. The most recent deep learning methods for spatio-temporal prediction are the main emphasis of this section.

7.5.1 Hybrid Spatio-temporal Deep Learning Models. In Hybrid spatio-temporal deep learning models, **Spatio-Temporal Graph Neural Networks (STGNNs)** have emerged as powerful tools for modeling complex spatial and temporal dependencies in various domains. A key element in Hybrid spatio-temporal deep learning models is the spatio-temporal fusion architecture, which combines the strengths of **Graph Neural Networks (GNNs)** or convolution neural networks for capturing spatial relationships with temporal models such as RNNs, LSTMs, or TCNs to model the evolution of data over time. Hybrid spatio-temporal deep learning model architectures can be broadly classified into factorized or coupled approaches. Factorized architectures separate spatial and temporal learning into distinct components, often using a GNN or CNN for spatial modeling and a separate temporal model, whose outputs are combined. Coupled architectures, however, integrate spatial and temporal learning more tightly within a single model. This fusion of spatial and temporal information enables Hybrid models to learn richer representations of the data, improving accuracy in predicting complex phenomena such as traffic flow, weather/climate patterns, crop patterns, and disease spread.

Convolutional-Recurrent Models (CRMs): CRMs are powerful neural network architectures designed to model spatial-temporal data concurrently in domains such as intelligent transportation, climate or weather prediction, agriculture yield prediction, and pandemic prediction. In CRMs, the convolutional layers handle spatial data, extracting local features and capturing spatial patterns from inputs such as road networks, satellite imagery, or sensor data in precision agriculture. The recurrent layers process the temporal sequence of the spatial features, capturing temporal dependencies and dynamics over time. This integration allows CRMs to effectively analyze spatio-temporal data, enabling accurate predictions and insights in complex real-world scenarios. For instance, in intelligent transportation, CRMs can predict traffic congestion patterns by leveraging road network structures and temporal traffic flow data. In climate and weather prediction, CRMs can capture spatial patterns of meteorological variables while considering their temporal evolution for more accurate forecasting. In agriculture, CRMs can model crop yield predictions by combining spatial sensor data with historical temporal records. In pandemic prediction, CRMs can analyze spatio-temporal disease spread patterns to forecast disease outbreaks and evaluate the impact of intervention strategies. CRMs offer a comprehensive and data-driven approach to jointly model spatial and temporal correlations, facilitating informed decision-making and optimization in these domains. Models like ConvLSTMs [29] can handle both the temporal and spatial dimensions and are presented as an alternative to stacking various neural networks. The convolution operation enables the model to capture the spatial information while requiring fewer parameters than a fully connected layer, as is the case for a traditional LSTM. The LSTM component of the ConvLSTM catches the temporal auto-correlation in the data, while the convolution operator captures the local spatial features. Stacked ConvLSTMs were used to address the problem of ConvLSTMs, which was unable to model all spatial-temporal dependencies, since they can learn hierarchical representations of the raw data. However, the seasonality and temporal periodicity of the data must be accurately detected in the typical spatio-temporal field. To do this, Wang et al. [177] segmented the time series after using a deep Spatio-temporal ConvLSTM. However, successful implementation requires careful hyperparameter tuning, model optimization, and sufficient training data for each specific application. While CRMs can exploit spatial and temporal correlations in the data, the sequential nature of recurrent layers often leads to vanishing and exploding gradients during training, hindering their ability to effectively model

long-range dependencies. Moreover, the propagation of errors through the layers and the inherent lack of ability of CNNs to handle irregular data motivate the use of Graph-Recurrent Models.

Graph-Recurrent Models (GRMs): GRMs comprise two primary constituents: the GNN strata and the recurrent strata. Within the GNN layers, the model undertakes the processing of spatial data, depicted in the form of a graph, wherein nodes embody spatial entities (such as road intersections, geographic locations, and sensor nodes). At the same time, edges encapsulate the spatial interconnections that bind them [193]. The GNN layers employ graph convolutions to aggregate information from neighboring nodes, capturing spatial dependencies and patterns. The recurrent layers, typically implemented as LSTM or GRU, handle the temporal aspect of the data, processing sequential observations over time and capturing temporal dependencies and dynamics. The output from the GNN layers, representing the spatial features, is combined with the output from the recurrent layers, representing the temporal features, to model the joint spatial-temporal correlations. This integration allows GRMs to concurrently analyze complex interactions between spatial locations and their temporal variations, making them well-suited for tasks such as traffic flow prediction, weather forecasting, agriculture yield estimation, and pandemic spread modeling. The model parameters are learned through backpropagation and optimization techniques, and the entire process is iteratively repeated during training to minimize prediction errors. While GRMs are better at handling irregular input than CNNs are, one of the main drawbacks of these approaches is that they are unable to capture long-term dependencies of the node attributes. For GRMs to extract significant patterns and dynamics, information must be propagated across spatial and temporal dimensions. Additionally, irregular temporal intervals in spatio-temporal data may be present; these intervals are challenging for traditional recurrent models to account for. The relationships between spatial and temporal components, or how changes in one dimension may affect the characteristics of another, are also poorly represented by these models.

Graph-Convolution Models (GCMs): To address the aforementioned limitations, models like GCN-TCN, incorporating a full convolutional structure on the time axis instead of recurrent neural networks, have shown promising results. To forecast traffic conditions, Yu et al. [198] introduce an innovative deep learning architecture known as the **Spatio-Temporal Graph Convolutional Network (STGCN)**. The proposed framework comprises multiple cascaded **spatio-temporal (ST)** convolutional blocks. Each block comprises two spatial graph convolution layers interleaved between two temporally gated convolutional layers. The model exhibits superior training performance compared to contemporary approaches with fewer parameters, as demonstrated through the experiments. Zhang et al. [165] also proposed using convolutions on both the temporal and spatial axes. They provide a novel graph-based approach called **Structure Learning Convolution (SLC)** to extract spatial connections. SLC uses the structure information to model local and global patterns separately by explicitly incorporating it into the convolution operation. The SLC modules can use static and dynamic graph information, since they use structure learning convolutions. The model incorporates pseudo-3D convolution to further represent temporal dependence. One major limitation is their computational complexity, particularly with large graphs, leading to longer training times and memory requirements. Additionally, as the number of layers increases, these models tend to over-smooth the node features, potentially losing crucial spatial information. Constructing an accurate graph representation from spatial data can also be challenging, impacting the model's performance. Generalizing to unseen nodes poses further challenges. While graph convolution models hold promise, addressing these limitations is crucial to fully unleash their potential in spatial modeling applications.

7.5.2 Attention-based Models. Attention mechanisms in deep learning empower models to dynamically weigh the importance of different elements in spatial-temporal data. Spatial attention discerns influential locations for predictions, while temporal attention highlights relevant

past-timesteps for forecasting. For instance, in traffic prediction, spatial attention might prioritize nearby roads, and temporal attention might focus on rush-hour patterns. This approach offers benefits such as improved accuracy, adaptability to varying patterns, and interpretability by revealing which locations and timesteps the model deems important. However, challenges remain in computational complexity and fully understanding the model's decision-making process. In Reference [172], the attention model's operation is described in depth. Here, we explain some spatial-temporal attention-based models for prediction tasks in various domains.

Intelligent Transportation System: In intelligent transportation, attention mechanisms enable the modeling of spatio-temporal dependencies in traffic data [17, 65]. For instance, attention mechanisms can assign higher weights to recent timesteps that are more relevant to the current prediction when predicting traffic flow or travel times. This allows the model to capture short-term patterns and adjust predictions accordingly. Attention mechanisms can also focus on specific periods or days of the week that exhibit recurring traffic patterns, enabling the model to account for temporal fluctuations and seasonality in the data.

Climate or Weather Data Modeling: In climate data modeling, attention mechanisms facilitate modeling spatio-temporal dependencies in weather patterns and climate variables [199]. For instance, attention mechanisms can assign weights to different timesteps based on their relevance to the current prediction when predicting temperature or precipitation. The model can capture long-term trends, seasonal patterns, and short-term fluctuations by attending to the historical climate data at specific time intervals. This enables accurate climate prediction and aids in understanding the temporal dynamics of climate change.

Precision Agriculture: In precision agriculture, attention mechanisms are applied to model the spatio-temporal dependencies between environmental factors, crop growth stages, and agricultural management practices. By incorporating attention mechanisms into crop yield prediction or irrigation scheduling models, temporal patterns and dependencies can be effectively captured [137]. Attention mechanisms can focus on relevant timesteps when predicting crop growth or optimizing resource allocation, such as irrigation and fertilization. This ensures that the models account for the varying needs of crops at different stages and adjust recommendations accordingly.

Pandemic Prediction: Attention mechanisms are crucial in modeling spatio-temporal dependencies in pandemic prediction. By incorporating attention mechanisms into epidemic models, such as **SEIR (Susceptible, Exposed, Infected, Recovered)** models, they can capture infectious disease spread and transmission dynamics over time [104]. Attention mechanisms allow the model to focus on each prediction's most relevant timesteps, considering factors such as incubation periods, contact tracing, and intervention effectiveness. This aids in accurate forecasting, understanding the impact of temporal variations in disease transmission, and optimizing public health interventions.

7.5.3 Use of Transformers. The Transformer model is a powerful neural network architecture originally developed for natural language processing tasks, such as machine translation and language modeling. However, it has also been successfully applied to other domains, including computer vision and spatio-temporal forecasting. In the context of spatio-temporal forecasting, the Transformer model can be used to model the complex spatio-temporal interrelationships present in the data. The key idea behind the Transformer is to replace the traditional RNN architecture with a self-attention mechanism that allows the model to attend to different parts of the input sequence at each timestep [5]. The Transformer model consists of an encoder and a decoder. The encoder processes the input sequence and produces a sequence of hidden representations, while the decoder takes the encoder output and generates the final output sequence. Each layer of the encoder and decoder contains a multi-head self-attention mechanism. In spatio-temporal

forecasting, the input sequence can represent a 3D tensor with dimensions (time, origin, destination) for transportation systems. The Transformer model can be used to capture the spatial and temporal dependencies in the data by treating each timestep as a separate sequence of spatial features. This allows the model to attend to different parts of the input sequence at each timestep and capture the complex spatio-temporal relationships present in the data. In addition to the multi-head self-attention mechanism, the Transformer model also includes feed-forward neural networks and layer normalization, which help to improve the stability and convergence of the training process. Transformer models have undergone significant advancements in spatio-temporal data modeling, starting from the initial models to the current state-of-the-art approaches. We discuss an overview of the evolution of transformer models for spatio-temporal data modeling.

Spatial-Temporal Transformer (STTransformer): The STTransformer was one of the early transformer-based models for spatio-temporal data [194]. It extended the transformer architecture to capture spatial and temporal dependencies, allowing it to model spatial relationships between data points and temporal dynamics. STTransformer is effective in tasks such as agriculture yield prediction, pandemic prediction, climate or weather prediction [74, 200], traffic prediction [100, 142], video prediction, and action recognition.

Graph Transformer: Graph Transformer introduces the concept of incorporating graph structures into transformer models for spatio-temporal data modeling [47, 128]. It considered the graph representation of data, capturing spatial relationships explicitly. Leveraging graph attention mechanisms enabled the model to attend to relevant graph nodes and aggregate information accordingly. Graph Transformer has been applied in tasks such as social network analysis, recommendation systems, and other spatio-temporal tasks [16, 98, 187].

Graph Convolutional Transformer (GCT): GCT combined the power of GCNs with transformers for spatio-temporal data modeling [69, 104]. It utilized graph convolutions to capture spatial relationships and transformers to model temporal dynamics. GCT effectively handled tasks such as spatio-temporal prediction in different domains, human activity recognition, and exploiting spatial and temporal dependencies in the data.

Video Transformer: Video Transformer specifically focused on modeling spatio-temporal dependencies in video data [23]. It extended transformer architectures to capture both spatial and temporal features in videos, leveraging self-attention mechanisms to model long-term dependencies across frames. Video Transformer achieved state-of-the-art performance in video classification, object detection, and video captioning tasks.

Domain-specific Transformers: In precision agriculture, techniques like **Crop Transformer (CropFormer)** [178] leverage transformer architecture tailored specifically for crop yield prediction. By incorporating domain knowledge about plant growth and environmental factors, CropFormer enhances prediction accuracy, aiding farmers in decision-making and resource allocation. Similarly, in the realm of weather and climate forecasting, transformer-based models such as MSAformer [180] and Climformer [24] have proven valuable. MSAformer excels at predicting $PM_{2.5}$ and other meteorological variables by employing attention mechanisms to focus on relevant atmospheric features. Climformer, however, is designed for long-term climate forecasting, utilizing self-attention to learn complex patterns from historical data and project future trends.

These are some of the notable advancements in transformer models for spatio-temporal data modeling. The field continues to evolve rapidly, with ongoing research focusing on improving model architectures, incorporating graph structures, handling long-range dependencies, and addressing the challenges specific to spatio-temporal data. In some recent state-of-the-art works, Amanambu et al. [5] proposed a model for forecasting hydrological drought using a deep transformer model. Li et al. [104] employ a Transformer to encode the time sequence information of COVID-19 and GCN to decode the time sequence information with graph structure. In a study, Liu

et al. [114] developed the transformer-based model Informer to forecast rice yield across the Indian Indo-Gangetic Plains by fusing time-series satellite data, environmental factors, and historical rice yield data from 2001 to 2016.

7.5.4 Generative Adversarial Network (GANs). GANs are a powerful deep learning technique that has been successfully applied to various tasks, including image and video generation. GANs can also be used for spatio-temporal data modeling including prediction tasks [179], where the goal is to generate realistic spatio-temporal data samples that capture the underlying dynamics of the system. In the context of spatio-temporal data modeling, GANs consist of two neural networks: a generator network and a discriminator network. The generator network takes as input a noise vector and generates a spatio-temporal data sample. The discriminator network inputs a spatio-temporal data sample and produces a binary classification output, indicating whether the sample is real or fake. Additionally, GANs can be used to perform data augmentation by generating new spatio-temporal data samples that are similar to the real data samples but with some variation. In the spatio-temporal prediction task, GANs forecast traffic flows in the TrafficGAN model [212]; CNN and LSTM models are included in TrafficGAN to capture the spatial-temporal correlations among the road connections of a road network. In another spatio-temporal task, to evaluate rainfall data for India and forecast future precipitation, Venkatesh et al. [173] created a rainfall prediction system employing GANs. However, training GANs for spatio-temporal data modeling can be challenging due to the complex spatio-temporal dependencies present in the data.

In Appendix B, we discuss recent techniques that can be applied with various models for capturing spatio-temporal dependencies in various domains.

8 Performance Study

Overall, deep learning techniques such as attention-based models and transformers generally excel across various problem types. For data structured in grid-like formations, CNNs are effective due to their ability to capture spatial dependencies. In contrast, attention-based models and transformers are well-suited for modeling spatial-temporal dependencies, because they can effectively capture long-range interactions, exploit global context, handle complex patterns, parallelize computations, and accommodate varying input sizes. (In Appendix C, Tables 9, 10, 11, and 12, present the performance results of various state-of-the-art models across different datasets in all domains.)

9 Conclusion and Future Directions

In conclusion, spatio-temporal forecasting is a rapidly evolving field that can potentially transform decision-making in many domains. With the increasing availability of data and the development of advanced statistical, machine learning, and deep learning techniques, the accuracy and scope of spatio-temporal forecasting are likely to continue to improve. Time series and statistical models are specifically designed to handle temporal patterns, making them effective for forecasting and trend analysis. They often require less data to train and can be computationally efficient. However, these models may struggle to capture complex spatial relationships or dependencies between variables in the data, limiting their performance for spatial-temporal tasks. In contrast, machine learning models offer flexibility and handle various data types. They can incorporate spatial features and perform well when the relationships between variables are not explicitly temporal, but traditional machine learning models may not fully exploit the temporal patterns in the data, leading to suboptimal performance for spatial-temporal tasks. Deep learning models, particularly attention-based models, excel at capturing complex spatial-temporal dependencies. They can automatically learn hierarchical features from the data, which is crucial for many spatio-temporal tasks. Deep learning models usually require a lot of data and computational resources to train effectively. They can

be challenging to interpret and may suffer from overfitting if not carefully regularized. Choosing the appropriate model for spatial-temporal data modeling depends on the specific task, available data, and computational resources. The future scope of spatio-temporal forecasting is vast, with potential applications in climate modeling, precision agriculture, smart cities, and public health. In the coming years, we can expect to see the development of more sophisticated models that can better capture complex spatio-temporal relationships in data. Additionally, integrating new data sources, such as satellite imagery and social media, will further enhance the accuracy and scope of spatio-temporal forecasting. However, some challenges must be addressed to fully realize the potential of spatio-temporal forecasting. Dealing with missing data, developing efficient algorithms, accounting for spatio-temporal dependencies, and incorporating uncertainty, scalability, and interpretability are a few of them. The interdisciplinary nature of spatio-temporal prediction opens up many future directions for research and innovation. Here are some potential areas for future development.

Integration of Machine Learning and Spatial Modeling: Integrating machine learning and spatial models enhances the quality of spatial-temporal prediction by combining machine learning's pattern recognition capabilities with the domain knowledge and spatial awareness obtained from the spatial models [91]. This integration enhances prediction accuracy, interpretability, and generalization while ensuring the predictions' physical plausibility. However, spatial data frequently involve complex data structures such as polygons, grids, or networks, posing challenges for seamless integration with machine learning models. Additionally, spatial data commonly exhibit spatial autocorrelation, where nearby locations are more similar than distant ones, thus challenging the independence assumptions inherent in many machine learning models.

Interpretable Models: Deep learning models excel at spatial-temporal prediction tasks, but their lack of interpretability hinders trust, bias detection, and improvement. While **Explainable AI (XAI)** techniques such as LIME and SHAP [143] can provide insights into individual predictions, they often struggle to accurately explain model predictions when the underlying relationships in the data are highly non-linear. Furthermore, the spatial autocorrelation in the data introduces additional challenges to the model's explainability.

Introducing Physical Constraints in Models: Recent advanced deep learning models excel in capturing intricate spatial-temporal relationships across diverse domains but frequently neglect practical real-world constraints in applications [106]. This oversight can hinder their adoption in professional settings. While recent strides in epidemic prediction demonstrate potential by integrating critical physical constraints such as differential equations into STGNN models, their applicability to other domains requires rigorous evaluation and testing.

Scalability: Deep learning models excel at spatial-temporal prediction tasks such as traffic forecasting and climate modeling, but their scalability suffers due to massive data volumes and complex architectures, hence suitable techniques for data compression, model pruning, and sparse attention need to be developed.

Different Modeling Techniques: Developing GNNs that dynamically adapt to changes in underlying graph structures is essential for capturing the dynamic essence of spatial-temporal data for spatial-temporal prediction tasks. Exploring models capable of handling heterogeneous graphs featuring node and edge types such as traffic sensors, road segments, and weather stations offers a more holistic representation of real-world systems. Furthermore, embracing the multi-view learning approach, which involves training distinct models on different data perspectives and then amalgamating their predictions, holds promise for enhancing the accuracy and robustness of models.

Overall, the interdisciplinary nature of spatio-temporal prediction opens up many opportunities for future research and innovation. By combining expertise from different fields, researchers can

improves generalization across different scenarios, requires less data to adapt to new tasks, and facilitates faster adaptation compared to training a model from scratch. This approach has found successful applications in traffic prediction, climate/weather forecasting, pandemic/disease modeling, and precision agriculture. Typically, meta-learning-based **Spatial-Temporal Graph Neural Networks (STGNNs)** involve extracting additional spatio-temporal attributes through a meta-learner. ST-MetaNet [129] is the pioneering study to introduce meta-learning into STGNNs, designed for spatial-temporal traffic prediction, which is composed of RNN, Meta-GAT, and Meta-RNN, and utilizes two types of meta-knowledge learners, namely, **Node Meta-Knowledge (NMK)** and **Edge Meta-Knowledge (EMK)** learners, to effectively incorporate additional spatio-temporal information. In Reference [147] Sao et al. propose MetaCitta, a novel deep meta-learning approach that addresses the critical challenges of data scarcity and model generalization and utilizes cross-correlations across tasks and cities. MetaCitta adopts the data from different cities and tasks in a generalizable spatio-temporal deep neural network. Yang et al. [195] introduce a novel **attention-based domain spatial-temporal meta-learning (ADST-ML)** model to predict $PM_{2.5}$ concentrations. This innovative approach utilizes historical $PM_{2.5}$ data, meteorological information, and temporal features to learn a strategy that can rapidly adapt to new prediction tasks in a given area. The model employs **convolutional neural networks (CNNs)** to extract spatial dependencies, an attention mechanism to focus on relevant information, and a **long short-term memory (LSTM)** network to capture temporal dynamics in $PM_{2.5}$ concentrations and other related factors. This allows the model to effectively predict the complex and often unobservable processes that influence changes in $PM_{2.5}$ levels. However, effective implementation requires careful consideration of task selection, model complexity, and data quality to ensure the model learns relevant patterns and generalizes well to new situations.

Transfer Learning leverages knowledge gained from a source task to improve performance on a related target task. For instance, a model trained on traffic data from one city can be adapted to predict traffic in another. This approach requires a large, well-labeled dataset for the source task and is ideal for adapting to new, but similar tasks. Meta-learning, however, focuses on learning to learn across a distribution of tasks. By training on various related tasks, the model learns a shared representation that captures commonalities, enabling rapid adaptation to new tasks even with limited data. This approach is well-suited for scenarios where the model needs to quickly adapt to new situations (or tasks) within that domain. Spatio-temporal graph transfer learning research can be broadly categorized into three approaches: clustering-based, domain adaptation-based, and meta-learning-based. Clustering-based approaches focus on grouping similar data points or regions to facilitate more efficient transfer learning by leveraging common cluster patterns. Domain adaptation-based approaches aim to reduce the discrepancy between the source and target domains, ensuring that the model trained on the source domain performs well on the target domain. Meta-learning-based approaches, however, seek to optimize the learning process itself, enabling the model to quickly adapt to new tasks with minimal data by learning to learn from previous experiences [1].

Self-Supervised Learning (SSL) is a promising approach for spatial-temporal prediction tasks. It leverages unlabeled data by creating pretext tasks encouraging models to learn meaningful spatio-temporal representations. For example, in traffic prediction, a model might be trained to predict future traffic flow from past data, learning about temporal dependencies and spatial correlations. In climate/weather prediction, the model could learn to fill in missing values or reconstruct future patterns from historical data. For pandemic modeling, it might predict disease spread based on data from other regions, and in precision agriculture, it could predict crop yields from sensor data and environmental conditions. This approach reduces the reliance on labeled data and enables the learning of complex representations that can significantly improve the

performance of prediction tasks. Contrastive learning, a prominent self-supervised learning method that utilizes positive and negative sample pairs, has recently been integrated into various graph neural networks-based deep learning models. A prime example is STGCL [113], pioneered by Liu et al., which marked the first successful incorporation of contrastive learning into a forecasting model. In Reference [161] Song et al. utilize contrastive learning on the CNN-LSTM model to construct an effective auxiliary task to learn feature representations of data in a self-supervised manner. Through these well-designed self-supervised tasks, uncomplicated networks obtain a strong capability to capture latent representations for weather changes with time-varying patterns.

Physics-Informed Learning (PIL) enhances spatial-temporal prediction by integrating physical laws and domain knowledge into machine and deep learning models. By incorporating physical constraints, such as traffic flow equations in traffic prediction [75], Navier-Stokes equations in weather prediction, epidemiological models in pandemic modeling, and plant growth models in precision agriculture, PIL ensures that predictions are physically plausible and consistent with real-world phenomena. This approach not only improves accuracy, particularly in data-limited scenarios but also enhances model generalization to unseen situations. However, it can increase model complexity and rely on the accuracy of the underlying physical models. Nonetheless, PIL offers a promising avenue for making more reliable and physically meaningful predictions in various spatial-temporal applications. Inspired by PINNs that are based on simple neural networks, physical-informed learning methods can also be combined with various deep learning models for spatial-temporal prediction tasks [206].

Federated Learning (FL) is emerging as a powerful tool for spatial-temporal prediction [103] across diverse fields such as traffic, climate/weather, agricultural produce, and disease/pandemic prediction. It enables collaborative model training while keeping sensitive data localized and addressing privacy concerns and data heterogeneity. FL allows transportation agencies and individuals to contribute their data in traffic prediction, leading to better traffic models and personalized recommendations [136]. For climate/weather prediction, meteorological agencies and sensor networks can collaborate to improve forecasts and climate models. In agriculture, farms and research institutions can share knowledge to optimize crop management and predict yields. Additionally, in the realm of pandemic and disease prediction, healthcare providers and public health agencies can work together for early detection and risk assessment. FL offers key benefits such as privacy preservation, data diversity, and reduced communication costs. However, challenges such as data and system heterogeneity, as well as privacy and security concerns, need to be addressed. Ongoing research focuses on personalized FL for tailored models, federated transfer learning to leverage knowledge from related tasks, and enhancing robustness and fairness in predictions. This approach holds immense potential for revolutionizing spatial-temporal prediction, leading to better decision-making and improved outcomes in various domains. In recent work on traffic prediction using federated learning, Liu et al. [111] perform the first study of forecasting traffic flow by adopting an **Online Learning (OL)** strategy in the FL framework and then propose a novel prediction method named **Online Spatio-Temporal Correlation-based Federated Learning (FedOSTC)**, aiming to guarantee performance gains regardless of traffic fluctuations. Later, Wen et al. [186] introduced a flexible distributed solar forecasting framework that combines a novel **spatial and temporal attention-based neural network (STANN)** with **federated learning (FL)**. This approach enables multi-horizon forecasting by training the model's feature extractor and forecaster on diverse local datasets for enhanced localization. The model's accuracy is further improved through global parameter aggregation without the need for centralizing data.

Online learning empowers spatial-temporal prediction models [111] to continuously learn and adapt as new data arrives, making it invaluable for dynamic environments such as traffic,

weather/climate, agriculture yield, and pandemic/disease forecasting. For instance, a traffic prediction model can ingest real-time sensor data, instantly updating its understanding of traffic flow and adjusting predictions to accommodate incidents or changing patterns. Similarly, in agriculture, weather forecasting, and disease prediction, online learning models dynamically incorporate the latest data to provide real-time, actionable insights for better decision-making. However, online learning is not without challenges. Models must grapple with concept drift, where data patterns shift over time, and catastrophic forgetting, where old knowledge is lost as new data is learned. Data quality is critical, as noisy or unreliable data can negatively impact model performance. Despite these hurdles, ongoing research into adaptive learning rates, forgetting mitigation, concept drift detection, and ensemble learning is paving the way for more robust and accurate online spatial-temporal prediction models. These advancements promise to unlock the full potential of online learning in tackling real-world problems where timely and accurate predictions are paramount.

C Results and Findings

In this section, we explore the industry applications of spatio-temporal research across various domains and compare cutting-edge models. We evaluate feature-based models and various deep learning models, including Convolution-RNNs, Graph-RNNs, Graph Convolution models, and Attention-based models, focusing on their performance potential in each domain. For evaluating the performance of models, accuracy is often viewed as the primary consideration when selecting a forecasting model; however, it is crucial to examine the tradeoffs between accuracy and other relevant factors. Here, we outline the typical performance metrics utilized in spatio-temporal forecasting to provide a comprehensive view of these considerations. The standard benchmark metrics used in existing literature in this domain include **Root Mean Squared Error (RMSE)**, **Mean Absolute Error (MAE)**, **Mean Absolute Percentage Error (MAPE)**, and **Coefficient of Determination (R^2)** [97]. The formulation of the mentioned metrics is defined below. Let $\mathcal{X}$ represent any spatio-temporal dataset. Assuming $X_{t,i} \in \mathcal{X}$ is the i th real-valued data point for a spatio-temporal dataset at timestep t , the corresponding predicted condition is $\hat{Y}_{t,i} \in \hat{\mathcal{Y}}_p$. Also, let $\bar{\mathcal{Y}}$ be the mean value of $\hat{\mathcal{Y}}_p$. Further, let N denote the number of nodes/regions/sensors in a particular network, and T_p the future prediction window. Then, the RMSE and MAE scores are given as:

$$RMSE = \sqrt{\frac{1}{N \times T_p} \sum_{t=1}^{T_p} \sum_{i=1}^N |X_{t,i} - \hat{Y}_{t,i}|^2}, \quad (1)$$

$$MAE = \frac{1}{N \times T_p} \sum_{t=1}^{T_p} \sum_{i=1}^N |X_{t,i} - \hat{Y}_{t,i}|. \quad (2)$$

Similarly, MAPE and Coefficient of Determination are defined as follows:

$$MAPE = \frac{1}{N \times T_p} \sum_{t=1}^{T_p} \sum_{i=1}^N \frac{|X_{t,i} - \hat{Y}_{t,i}|}{|X_{t,i}|}, \quad (3)$$

$$Coefficient_of_Determination(R^2) = 1 - \frac{\frac{1}{N \times T_p} \sum_{t=1}^{T_p} \sum_{i=1}^N |X_{t,i} - \hat{Y}_{t,i}|^2}{\frac{1}{N \times T_p} \sum_{t=1}^{T_p} \sum_{i=1}^N |X_{t,i} - \bar{\mathcal{Y}}|^2}. \quad (4)$$

Table 9. Prediction Performances of Traffic Forecasting Models on Multiple Datasets

Model	METR-LA			PEMS-BAY			PEMS04			PEMS08		
	15 min	30 min	60 min	15 min	30 min	60 min	15 min	30 min	60 min	15 min	30 min	60 min
Conv-LSTM	5.6	6.3	7.3	4.2	2.8	5.6	31.5	33.4	35.2	24.1	25.6	27.0
	3.5	4.7	5.4	2.1	3.3	5.6	20.5	21.7	23.6	16.3	18.2	20.7
	8.2	9.1	10.4	3.2	4.8	5.6	14.6	16.5	17.9	11.3	12.9	13.8
STGCN	4.8	5.5	6.4	2.5	3.0	3.6	29.5	30.5	31.9	22.9	23.9	25.3
	2.6	2.8	3.1	1.3	1.5	1.7	18.7	19.3	20.1	14.8	15.3	16.2
	6.6	7.4	8.5	2.8	3.2	3.9	13.1	13.4	13.9	9.81	10.1	10.6
Graph WaveNet	4.8	5.5	6.3	2.2	2.8	3.5	27.7	28.6	29.8	20.7	22.0	23
	2.6	2.8	3.1	1.1	1.3	1.6	17.2	17.8	18.6	13.0	13.6	14.3
	6.6	7.6	8.9	2.2	2.8	3.5	12.0	12.5	13.4	8.8	9.0	9.8
DDSTGCN	5.0	6.0	7.1	2.71	3.7	4.4	28.4	29.6	30.69	21.1	22.2	24.1
	2.7	3.0	3.4	1.3	1.6	1.9	17.7	18.5	19.5	13.6	14.0	14.7
	6.8	8.1	9.8	2.7	3.6	4.5	12.3	12.7	13.5	8.8	9.7	10.4
LSTTN	4.4	5.1	5.9	2.2	2.8	3.4	27.7	28.7	30.0	20.9	21.9	23.5
	2.4	2.7	2.9	1.06	1.3	1.5	17.1	17.6	18.5	13.2	13.5	14.6
	5.9	6.8	7.9	2.1	2.6	3.2	11.3	11.7	12.2	8.6	9.0	9.7

The best results are marked in **bold**. Each row displays the RMSE, MAE, and MAPE values corresponding to the model's performance.

The coefficient of determination, R^2 , measures the goodness of fit of a model. R^2 is a statistical measure of the percentage variation in X , which is explained by the model. An R^2 nearer to 1 indicates that the model can suitably fit the data. Now, let us compare different cutting-edge models and the industry applications of research on spatial-temporal predictions in various domains.

Intelligent Transportation System: Deep learning techniques have become prominent in **Intelligent Transportation Systems (ITS)**, particularly for spatial-temporal traffic prediction, owing to their ability to capture complex traffic patterns effectively. GNNs and their variants, incorporating recurrent or attention mechanisms, stand out among these techniques, addressing spatial-temporal dependencies more efficiently than conventional methods. Industries leveraging ITS applications include dynamic traffic light control, smart routing for navigation systems, automated collision avoidance, and environmentally focused driving algorithms, collectively improving urban mobility and promoting sustainable urban planning by adapting to real-time conditions and predicting future transportation needs [72]. Here, we compare state-of-the-art models across four real-world traffic datasets for various prediction tasks (flow/volume/speed), with results summarized in Table 9. The models include Conv-LSTM [29], STGCN [198], Graph WaveNet [192], **Dual dynamic spatial-temporal graph convolution network for traffic prediction (DDSTGCN)** [164], Transformer network with decoupled spatial-temporal embedding for traffic flow forecasting, **LSTTN: A Long-Short Term Transformer-based Spatio-temporal Neural Network for Traffic Flow Forecasting** [116]. Our research reveals that the LSTTN model consistently achieves high prediction accuracy across various datasets. While it may show slightly lower performance on the PEMS08 dataset than Graph WaveNet, LSTTN significantly outperforms baseline models such as METR-LA and PEMS-BAY. For instance, on METR-LA, LSTTN outperforms all baseline models across all metrics and prediction horizons. Overall, these findings emphasize the importance of attention mechanisms and transformer networks, which incorporate both long-term trend analysis and periodic patterns in spatial-temporal

Table 10. Performance Comparison of Different Approaches for Weather Forecasting on Four Benchmark Datasets

Models	Metrics	Beijing			Israel				USA			
		Tem	Hum	Ws	Tem	Hum	Ws	Ps	Tem	Hum	Ws	Ps
XGBoost	RMSE	2.66	14.70	1.24	1.51	10.31	1.22	4.31	2.27	12.53	1.67	4.49
	MAE	1.97	10.76	0.89	1.06	7.19	0.84	2.33	1.62	9.13	1.23	2.16
LSTNet	RMSE	2.66	14.69	1.26	1.64	10.86	1.28	4.49	2.43	12.63	1.79	4.19
	MAE	2.02	10.49	0.88	1.19	7.65	0.91	2.41	1.76	9.12	1.28	2.19
AGCRN	RMSE	2.48	14.97	1.27	1.65	11.18	1.31	4.32	2.39	12.39	1.90	4.43
	MAE	1.89	10.50	0.86	1.21	7.71	0.95	2.56	1.76	9.03	1.36	2.61
Transformer	RMSE	2.76	15.49	1.27	1.73	11.32	1.31	5.32	2.33	12.22	1.79	3.92
	MAE	2.08	10.96	0.92	1.25	7.71	0.93	3.22	1.66	8.76	1.28	1.877
STCWF	RMSE	2.52	13.64	1.23	1.53	10.29	1.245	3.93	2.16	12.12	1.69	3.84
	MAE	1.91	9.66	0.85	1.09	7.03	0.87	2.14	1.545	8.69	1.23	1.91

Tem, Hum, Ws, and Ps denote temperature, humidity, wind speed, and pressure, respectively. The best results are in **bold**.

traffic prediction tasks, driving the continuous evolution of deep learning techniques in this domain.

Weather and Climate: The practical applications of advancement in climate/weather data modeling span several industries [127]. In agriculture, improved forecasts can inform decisions from planting to harvesting, optimizing resources, and increasing yields. The energy sector, particularly renewable energy, relies on precise weather predictions for energy generation and grid management. Aviation and maritime industries use these forecasts for routing and safety, while urban planning utilizes them for infrastructure resilience against extreme weather. We assessed several state-of-the-art machine and deep learning models, including XGBoost [28], LSTnet [99], AGCRN [11], transformer [172], and STCWF [56], on various weather datasets, such as Beijing, Israel, and USA [56], for climate/weather forecasting. The performance of all models is shown in Table 10. STCWF consistently outperforms other approaches across multiple datasets, demonstrating significant improvements in RMSE and MAE, particularly on the Beijing dataset. XGBoost shows strong performance, especially on the Israel and USA datasets, underscoring the efficacy of ensemble algorithms, while models like AGCRN overlook inter-variable influence. STCWF, leveraging self-supervised learning, excels across various datasets, outperforming other baselines, and emphasizing its superiority in weather forecasting.

Precision Agriculture: In the quest to enhance the accuracy of crop yield predictions, several machine learning and deep learning-based models have been deployed for spatial-temporal crop yield prediction in precision agriculture. Frontier research and practical applications in various precision agricultural datasets are pivotal for enhancing agricultural productivity and sustainability. Researchers employ advanced techniques, particularly deep learning, remote sensing data fusion, and geostatistical analysis, to develop crop yield prediction models, monitor crop health, assess soil quality, and analyze agricultural census trends [154]. We compare (Table 11) performance of various machine learning models such as XGBoost [140] and state-of-art deep learning models, CNN-RNN [89], GNN-RNN [193], and CNN-LSTM-Attention [115] on various crop yield datasets of Maize, Rice, and Soybean in China for forecasting crop yield. Models were trained and tested using a comprehensive dataset encompassing Maize, Rice, and Soybean. Among these, the CNN-LSTM-Attention model outperformed others. This model, integrating spatial and temporal processing with an attention mechanism, demonstrated significant improvements over

Table 11. Training and Testing Performance Metrics for Different Crop Models

Crop Type	Model	Training Dataset			Testing Dataset		
		R ²	RMSE (kg/ha)	MAPE (%)	R ²	RMSE (kg/ha)	MAPE (%)
Maize	XGBoost	0.74	400.18	4.30%	0.70	457.04	4.92%
	CNN-RNN	0.78	350.12	3.90%	0.73	394.67	4.30%
	GNN-RNN	0.81	332.25	3.77%	0.78	385.09	4.18%
	CNN-LSTM-Attention	0.85	320.45	3.70%	0.80	375.08	4.21%
Rice	XGBoost	0.72	580.24	5.80%	0.67	612.10	6.12%
	CNN-RNN	0.75	550.12	5.40%	0.70	585.40	5.59%
	GNN-RNN	0.78	535.30	5.20%	0.72	558.52	5.38%
	CNN-LSTM-Attention	0.80	510.05	5.00%	0.76	519.07	4.67%
Soybean	XGBoost	0.62	240.10	8.80%	0.57	264.88	9.58%
	CNN-RNN	0.75	200.15	7.00%	0.73	210.96	7.13%
	GNN-RNN	0.78	191.20	6.70%	0.75	198.07	6.88%
	CNN-LSTM-Attention	0.80	180.10	6.50%	0.78	188.11	6.39%

Table 12. Performance Comparison of Different Models on the Various US Pandemic Datasets for T = 5 days and T = 10 Days

Model	T = 5 days			T = 10 days		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE
ARIMA	6,475	22,095	14.01%	11,489	44,779	26.36%
CNN-LSTM	18,348	32,950	21.88%	26,749	47,328	32.52%
GraphWaveNet	13,875	22,559	17.85%	9,526	15,673	16.64%
CovidGNN	9,453	21,612	9.91%	16,052	37,586	15.03%
PatchTST	5,086	12,119	7.62%	8,033	17,283	11.70%
MPSTAN	3,960	8,255	6.38%	7,711	14,463	10.73%

traditional methods like XGBoost and other deep learning approaches such as CNN-RNN and GNN-RNN.

Pandemic and Disease: Practical applications of spatial-temporal pandemic and disease prediction are widespread across industries [121]. Healthcare systems implement real-time surveillance and diagnostic AI to manage resources and detect diseases early. Public health authorities utilize contact tracing apps and spatial modeling for effective vaccination distribution and outbreak management. In travel and mobility, predictive models assess risks and inform dynamic advisories to prevent disease transmission. Deep learning techniques are emerged as an advanced method to model pandemic and disease data. We evaluated the performance of various traditional models such as ARIMA [57], and state-of-the-art deep learning models, including Graph Wavenet [192], CovidGNN [85], PatchTST [123], and MPSTAN [118] on Japanese and USA pandemic datasets [37] to predict future disease spread over the regions. The performance of all models is shown in Table 12 for the USA dataset. Our evaluation of the USA dataset reveals that the attention-based model MPSTAN’s state-of-the-art performance in predicting disease spread over 5 and 10 days surpasses other models in MAE, RMSE, and MAPE. While traditional models like ARIMA excel in short-term forecasting, they falter with longer forecasts. PathTST is also a transformer-based model with good results compared to other deep learning models, showing the supremacy of attention-based models. MPSTAN’s success underscores the efficacy of integrating epidemiological domain knowledge into deep learning frameworks, particularly in meta-population transmission modeling.

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